

Lemma

What we estimate when we estimate dynamic causal effects in panel data

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Abstract

Policy-relevant decisions depend on causal effects in the current or next period (e.g., FATT, FATU), but standard panel-data methods target backward-looking estimands such as the average treatment effect on the treated (ATT) or group-time ATTs over the observed window. We formalize this gap using future causal effects and present three identification strategies. Taking dynamic effects *less* seriously, the FATT is identified under time homogeneity by the ATT or by a convex combination of within-group ATTs (Path 1). Taking them *more* seriously, the FATT is identified either by specifying how group-time effects evolve and extrapolating (Path 2), or by modeling structural heterogeneity via covariates and integrating over the future covariate distribution (Path 3). Path 3 enables identification under regime change when reduced-form temporal patterns break but structural covariates remain predictive. For Path 2, we provide the first systematic framework for model selection in causal extrapolation, using time-series cross-validation to test extrapolation performance rather than in-sample fit. We give efficient influence functions for the FATT under all three paths so that standard errors and confidence intervals propagate correctly from first-stage estimators.

1 Introduction

There is a fundamental and unacknowledged tension at the heart of policy analysis and causal inference with panel data more broadly [Egami and Hartman, 2023, Heckman and Vytlačil, 2001]. This tension arises because of the possibility of dynamic causal effects (effects that change over time), which, as we'll argue below, preclude an important type of external validity: if policy effects are truly unrestrictedly dynamic, then it is impossible to use historical data to inform *any* current decisions, which decisions are presumably the reason we care about estimating any causal effects in the first place.

Most causal inference rests on an implicit assumption of time homogeneity. We might call this assumption the Fundamental Promise of Causal Inference: causal effects estimated (by necessity) on data collected in the past can be used to inform decision-making in the present [Egami and Hartman, 2023, Khosrowi, 2019, Cartwright, 2007]. If two identical clinical trials – same target population and inclusion criteria, same treatment, same outcome – were expected to not only give different

results but in fact target different causal effects simply because of the passage of time, that would be a major crisis for causal inference. A durable understanding of causal effects would be elusive.

Public policies are widely understood to violate this assumption and are assumed to have effects that inherently change over time [Lucas Jr, 1976, Ericsson et al., 1998, Estrella and Fuhrer, 1999]. Policies are introduced into dynamic contexts, may be administered or enforced differently over time, and may themselves cause changes in public behavior and attitude. Technological advancement, public opinion, economic conditions, and new research may all modify the causal effects policies.

This basic assumption of dynamic effects has driven a great deal of recent advances in methods for causal inference with panel data [Sun and Abraham, 2021, Wang et al., 2024]. Recent surveys clarify how heterogeneity and dynamics enter identification and estimation in this setting [Arkhangelsky and Imbens, 2024, De Chaisemartin and d'Haultfoeuille, 2023]. It is now standard to assume that causal effects vary both with time of treatment and time since implementation. These effects (often called “group-time” effects) may be of interest in themselves but more typically are aggregated into coarser effects like an “overall” average treatment effect on the treated (ATT). Assuming such dynamic effects solves a problem with classical methods like two-way fixed effects: they may incur bias in the presence of time-heterogeneous effects because they implicitly include comparisons between units treated at different times in their estimate of the ATT [Sun and Abraham, 2021].

While methods assuming dynamic effects have been developed for good reasons, they strike at the heart of the Fundamental Promise of Causal Inference. In the presence of dynamic effects, it is not clear how one can connect causal effects estimated on historical data to any decision one would make now or in the future [Egami and Hartman, 2023, Devaux and Egami, 2022, Khosrowi, 2019]. The question of when and how historical evidence can inform current or future decisions is central to the literature on generalizing and transporting causal effects [Cole and Stuart, 2010, Tipton, 2013, Manski, 2013]. If one estimates the effect of a public policy to be beneficial (e.g., improves life expectancy) among states that implemented it in historical data, does that mean that it would be wise to maintain that policy in places that have already implemented and/or to implement it in places it has not been implemented? If effects are unrestrictedly dynamic, it’s impossible to say.

We seek to resolve this tension between dynamic effects and the Fundamental Promise by encouraging researchers to either take the possibility of dynamic effects *less* seriously or *more* seriously—and, when taking them more seriously, to choose what they believe is invariant. Taking it less seriously (Path 1) involves assuming some level of time homogeneity in causal effects. This would allow causal effects that would inform current and future decisions to be identified in historical data. We argue that an informal version of this is currently standard practice, where average treatment effects estimated on historical data are treated as if they are stable enough over time to inform current debates [Gabler et al., 2009, Cintron et al., 2022]. The notion of policy-relevant treatment effects formalizes which estimands answer counterfactual policy questions [Heckman and Vytlacil, 2001, Sasaki and Ura, 2023]. On the other hand, taking dynamic effects more seriously would require formally modeling how effects change over time. Under Path 2, researchers specify temporal dynamics (e.g., trends in event time) and extrapolate [Khosrowi, 2019, Maeba, 2024]. Under Path 3, researchers model structural heterogeneity via covariates and integrate over the future covariate distribution. Path 3 is robust to regime change if treatment effects are driven by observable

covariates that capture “deep parameters” [Lucas Jr, 1976]—even when reduced-form temporal patterns break, structural covariates may remain predictive. This approach connects to structural methods that enable extrapolation across regimes [Heckman and Vytlačil, 2005, Pearl and Bareinboim, 2022].

We introduce new causal estimands that formalize the notion that causal inference is inherently future-oriented, which we call “future causal effects.” We locate the estimands outside the time range of the current study and present three identification paths that differ in what they assume is invariant: Path 1 assumes temporal stability of effects; Path 2 assumes a functional form for temporal dynamics; Path 3 assumes structural covariates remain predictive under regime change. Across fields, three broad strategies emerge for making causal claims about the future: structural or parameter invariance (e.g., policy-invariant deep parameters); statistical transport or reweighting of effects across populations; and explicit modeling of heterogeneity or dynamics. Our three paths map cleanly onto these traditions, with Path 3 most directly invoking the Lucas critique’s emphasis on structural parameters.

1.1 Related work

Recent advances in causal inference for panel data and difference-in-differences provide the backbone for our setting. Surveys by Arkhangelsky and Imbens [2024] and De Chaisemartin and d’Haultfoeuille [2023] clarify how heterogeneity and dynamics enter identification and estimation; the group-time and staggered-adoption literature [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021, Roth and Sant’Anna, 2023, Baker et al., 2025] addresses inference for backward-looking effects when treatment timing varies. Work on testing parallel trends and time homogeneity—including the limitations of conventional pre-trend tests [Freyaldenhoven et al., 2019], the pitfalls of conditioning on passing such tests [Roth, 2022], inference under bounded trend violations [Rambachan and Roth, 2023], and equivalence tests for pre-trends [Dette and Schumann, 2024]—connects directly to the testable implications of our time-homogeneity assumptions. These lines of work focus on identification and inference for effects in the observed window; we instead target estimands that are relevant for current or future policy decisions.

The disconnect between typical estimands (ATT, group-time effects) and policy-relevant or future causal questions has been formalized in several ways. Heckman and Vytlačil [2001] define policy-relevant treatment effects for counterfactual policy changes; Sasaki and Ura [2023] provide estimation and inference for such parameters. Forastiere et al. [2025] formalize forecasting the causal effect of *re-implementing* an intervention at a future time; we focus on the effect of the policy in the current or next period ($p + 1$)—continuing or repealing for current adopters, adopting or not for non-adopters—and potentially beyond. Gische et al. [2021] distinguish forecasting the causal effect of an intervention from predicting future outcomes; we similarly distinguish backward-looking estimands (e.g., the ATT in the observed window) from forward-looking, policy-relevant estimands such as the FATT and FATU at time $p + 1$.

External validity and transportability are central to our concern that backward-looking estimands may not inform current

decisions. [Egami and Hartman \[2023\]](#) provide a framework for design and analysis; [Pearl and Bareinboim \[2022\]](#) and [Bareinboim and Pearl \[2012\]](#) give graphical and identification results for transporting causal effects. The literature on generalizing from randomized trials to target populations [[Cole and Stuart, 2010](#), [Tipton, 2013](#)] and on robustness to external validity bias [[Devaux and Egami, 2022](#)] is closely related. [Khosrowi \[2019\]](#) and [Khosrowi \[2023\]](#) examine the assumptions and uncertainties underlying extrapolation; [Manski \[2013\]](#) advocates bounded identification when extrapolation is weak. [Shpitser \[2013\]](#) extend transportability to longitudinal settings.

The Lucas Critique and the question of policy invariance underlie our discussion of when effects can be taken as stable over time. [Lucas Jr \[1976\]](#) argues that policy evaluation requires parameters that are invariant to the policy regime; [Ericsson et al. \[1998\]](#) and [Estrella and Fuhrer \[1999\]](#) formalize and test such invariance. Structural and parametric approaches to extrapolation—including [Heckman and Vytalil \[2005\]](#) on policy-invariant parameters, [Pearl and Bareinboim \[2022\]](#) on transportability, and [Brodersen et al. \[2015\]](#) on Bayesian structural time series—show how functional assumptions enable extrapolation across regimes or time periods.

Philosophical work on causal invariance and extrapolation [[Cartwright, 2007](#)] and empirical evidence on heterogeneous treatment effects in policy and program evaluation [[Gabler et al., 2009](#), [Cintron et al., 2022](#)] further motivate the need to be explicit about which effects are policy-relevant and under what assumptions they can be identified from historical data.

Our future-looking estimands sit at the intersection of three strategies for making causal claims about the future: structural or parameter invariance, statistical transport or reweighting, and explicit modeling of heterogeneity or dynamics.

2 Setting and context

Suppose we measure an indicator of a policy A_{it} , a possibly affected outcome Y_{it} , and time-invariant covariates X_i for $i = 1, \dots, n$ units (often the units are geographic entities, like states, counties, etc.) and $t = 1, \dots, p$ timepoints. For ease of exposition, we take A_{it} to be a policy, but all that follows applies equally to other types of interventions. Define the counterfactual value that Y_{it} would take if the policy were set, possibly contrary to fact, to $A_{it} = a$ as $Y_{it}(a)$. We assume no interference: each unit’s potential outcomes depend only on that unit’s treatment, so $Y_{it}(a)$ is well-defined. Let $\mathcal{O} = \{X_i, Y_i(0), Y_i(1), A_i\}_{i=1, \dots, n} \sim \mathbb{P}$ where $Y_i(a) = \{Y_{it}(a)\}_{t=1, \dots, p}$ and $A_i = (A_{it})_{t=1, \dots, p}$. We extend the setting to future time periods when defining future estimands below. The distribution of data at time t is $\mathbb{P}_t$. For integer $k \geq 1$, we write $\mathbb{P}_{p+k}$ for the distribution of $(X_i, Y_{i,p+k}(0), Y_{i,p+k}(1), A_{i,p+k})$ in the same superpopulation k periods ahead (i.e., the marginal at $t = p + k$ of the joint distribution governing the process through $p + k$); $\mathbb{P}_{p+1}$ is the case $k = 1$.

It is typical (though not necessary) in such studies to take the target of estimation to be the average treatment effect on the treated (ATT). The ATT may be defined as

$$\theta = \mathbb{E}_{\mathbb{P}}\{Y_{it}(1) - Y_{it}(0) | A_{it} = 1\}.$$

One virtue of the ATT is that $Y_{it}(1)$ is always observed under the conditioning event $A_{it} = 1$, so identification of the ATT is facilitated by only requiring causal inference machinery to estimate $\mathbb{E}_{\mathbb{P}}\{Y_{it}(0)|A_{it} = 1\}$.

In the presence of heterogeneity – especially heterogeneity in how policy effects change over time – there have been proposals to instead decompose the ATT into separate sub-ATTs, with an example being referred to as group-time ATTs [Sun and Abraham, 2021].

$$\theta_{gt} = \mathbb{E}_{\mathbb{P}_t}\{Y_{it}(1) - Y_{it}(0)|A_{it} = 1, G_i = g\}$$

where $G_i = \min_t tI\{A_{it} = 1\}$, the time when the policy first occurred, and $\mathbb{P}_t$ is the distribution of the data at time t : $\mathcal{O}_t = \{X_i, Y_{it}(0), Y_{it}(1), A_{it}\}_{i=1,\dots,n} \sim \mathbb{P}_t$. This is important to solve some infelicities with trying to estimate θ directly with a single linear model [Sun and Abraham, 2021] and to draw attention to the fact that causal effects may both vary across calendar time and have different effects as implementation continues – i.e., effects may differ in the first year of implementation from the fifth year, say. These sub-ATTs are typically then aggregated up into an overall effect estimate across groups/times.

2.1 Identification of the backward-looking ATT

The results in Section 4.1 (covering all three identification paths) assume that the researcher has secured identification of the ATT θ and, where relevant, the group-time ATTs θ_{gt} , under some design. For example, under parallel trends in difference-in-differences, θ_{gt} is identified from the observed data [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021]; under unconfoundedness or other designs, different identification formulas apply. We do not restrict the first-stage identification strategy: it may be difference-in-differences, synthetic control, or another approach. The contribution of this paper is to show when and how these backward-looking estimands coincide with or can be used to identify the policy-relevant FATT (and related future estimands), given time homogeneity assumptions (Path 1, Section 4.2), a parametric model for dynamics (Path 2, Section 4.5), or structural covariate-driven effects (Path 3, Section 4.6). Thus the full identification argument has two steps: (a) under the researcher’s design, θ or θ_{gt} are identified from observables; (b) under the extrapolation assumptions we state below, the FATT is identified from θ or θ_{gt} .

However, regardless of how they are aggregated, these effects are typically not directly relevant to current policy or decisionmaking [Egami and Hartman, 2023, Khosrowi, 2019]. These ATTs necessarily correspond to what has happened *in the past*. While assessing the effects of policies in the past may have some utility – e.g., for electoral purposes – for a decisionmaker considering implementing a new policy or repealing a policy that is currently on the books, these backward-looking estimands are not germane without further assumptions.

3 Estimands

We require more policy-relevant estimands that are future-looking and correspond to “future causal effects.” These estimands are designed to address actual policy decisions: should we implement or repeal this policy? We define them for a generic horizon $k \geq 1$ (i.e., k periods ahead); conditioning on treatment status is at the end of the study period p (e.g., effect for current adopters or non-adopters at horizon k).

Let

$$\tau_{i,p+k} = Y_{i,p+k}(1) - Y_{i,p+k}(0)$$

be the future individual treatment effect (FITE) for unit i at horizon k . The FITE is arguably the most relevant quantity for a citizen or policymaker belonging to unit i . For example, state lawmakers in California may likely be interested most in the FITE for California. The fact that individual treatment effects are not as a rule identifiable does not change the fact that it may be what is of primary importance. Do note also that none of the future causal effects we will discuss in this section are identifiable from the observed data without further assumptions.

Given the difficulties with estimating individual treatment effects, we may consider an alternative, the future average treatment effect among similar units (FATS). Given a distance measure ρ , we can define:

$$\tau_{i,p+k}(\eta; \rho) = \mathbb{E}_{\mathbb{P}_{p+k}} \{Y_{j,p+k}(1) - Y_{j,p+k}(0) | \rho(X_i, X_j) \leq \eta\}$$

i.e., the average treatment effect at $p+k$ among units j such that $\rho(X_i, X_j) \leq \eta$; similarity is defined by distance in covariate space. (The backward-looking analog is the ATS: the average effect among units in that same set of units, under $\mathbb{P}_t$.) For example, policymakers in Massachusetts may be interested in the treatment effect among states in New England, or New York policymakers may be interested in the effect among other high-population states.

When policymakers do not have a unit-specific focus, other treatment effects may be of interest. Consider the future-looking analogs of typical backward-looking estimands at horizon k :

$$\tau_{p+k} = \mathbb{E}_{\mathbb{P}_{p+k}} \{Y_{i,p+k}(1) - Y_{i,p+k}(0)\} \quad (FATE)$$

$$\theta_{p+k} = \mathbb{E}_{\mathbb{P}_{p+k}} \{Y_{i,p+k}(1) - Y_{i,p+k}(0) | A_{ip} = 1\} \quad (FATT)$$

$$\omega_{p+k} = \mathbb{E}_{\mathbb{P}_{p+k}} \{Y_{i,p+k}(1) - Y_{i,p+k}(0) | A_{ip} = 0\} \quad (FATU)$$

The FATE encodes the average outcome at time $p+k$ if all units implemented or continued the policy compared to if all units discontinued or remained without the policy. The FATE is similar to the average treatment effect (ATE), and considers the

most extreme counterfactual scenario of all implementing units versus none. The FATT encodes the difference in average outcome for units with the policy implemented at time p (the end of the study period) if they continued the policy compared to if they discontinued the policy at time $p + k$. The FATT is analogous to the ATT and is targeted at the effect of repealing the policy. The FATU similarly encodes the difference in average outcome among units without the policy at time p if they newly implemented the policy compared to if they remained without at time $p + k$. The FATU is most relevant to policymakers considering whether they should implement the policy.

Unlike the ATT and sub-ATTs above, none of the counterfactuals are observed. In fact, unlike for any backward-looking estimand, we lack any direct knowledge of the distribution of $\mathbb{P}_{p+k}$. We therefore require additional machinery to connect the observed data to the policy-relevant distribution $\mathbb{P}_{p+k}$. We outline below two different options: either taking time heterogeneity *less* seriously – by making time homogeneity assumptions – or taking it *more* seriously and formally modeling how policy effects change. **In the remainder of the paper we take $k = 1$ (one period ahead)**; extension to $k > 1$ is analogous and noted where it differs.

4 Identification via Extrapolation Functions

Forward-looking estimands like the FATT at $p + 1$ require extrapolation beyond the observed data. The three identification paths we present can be unified under a single conceptual framework: each path specifies an *extrapolation function* f and assumes that some structural relationship is invariant across time. What differs across paths is the choice of f —what it depends on and what is assumed stable.

4.1 General framework

All three paths share a common principle: they express the group-time treatment effects θ_{gt} as a function of parameters and “information” available at time t , then extrapolate this relationship to $t = p + 1$. Formally, we write

$$\theta_{gt} = f(g, \mathcal{I}_t; \boldsymbol{\theta}), \quad t = 1, \dots, p,$$

where $f(\cdot)$ is an extrapolation function, $\mathcal{I}_t$ represents available information at time t , and $\boldsymbol{\theta}$ are parameters identified from the observed $\{\theta_{gt}\}_{t \leq p, g \in \mathcal{G}}$. The FATT is then

$$\theta_{p+1} = \sum_{g \in \mathcal{G}} \omega_g f(g, \mathcal{I}_{p+1}; \boldsymbol{\theta}),$$

where $\omega_g = \mathbb{P}(G_i = g | A_{ip} = 1)$ is the probability of cohort g among units treated by period p .

The three paths differ in their specification of f and $\mathcal{I}_t$:

Table 1: Three identification paths as choices of extrapolation function

Path	Function f	Input $\mathcal{J}_t$	What's Invariant
1	γ_g (constant in t)	Group index only	Group-specific effects
2	$f(g, t; \gamma)$	(Group, Time)	Temporal functional form
3	$\int m(x; \beta) dF_X^t(x)$	Covariate dist. at t	Conditional effect $m(x; \beta)$

- **Path 1 (time homogeneity):** f is constant over time within each group; $\mathcal{J}_t$ contains only the group index g . This assumes group-specific treatment effects do not change over time.
- **Path 2 (parametric extrapolation):** f is a parametric function of group and time (or event time $k = t - g$); $\mathcal{J}_t$ encodes temporal dynamics. This assumes effects follow a smooth functional form that continues into the future.
- **Path 3 (covariate integration):** f depends on the covariate distribution F_X^t at time t ; $\mathcal{J}_t$ is the population composition. This assumes the conditional effect $\tau(x) = m(x; \beta)$ is invariant, even if the covariate distribution shifts.

General identification principle. Identification of θ_{p+1} requires: (i) the extrapolation function f is correctly specified; (ii) the structural relationship encoded in f holds at $t = p + 1$ (extrapolation validity); (iii) the parameters θ are identified from observed data $\{\theta_{gt}\}_{t \leq p}$. When these conditions hold, plugging in the estimated $\hat{\theta}$ and known or forecasted $\mathcal{J}_{p+1}$ yields an identified estimator of the FATT. The choice among paths should be guided by which structural assumption—time homogeneity, temporal stability, or covariate-conditional invariance—is most credible in the application at hand.

We now present each path in detail, stating the choice of f , the required assumptions, and the identification result.

4.2 Path 1: Time-homogeneous functions

There is such a rich and growing literature on estimation of the ATT and sub-ATTs [Wang et al., 2024, Sun and Abraham, 2021] that researchers may be interested in still using these tools to estimate forward-looking, policy-relevant estimands. Here, we outline in what cases this may be justified. We take as given that the researcher’s design (e.g., parallel trends in DiD) has secured identification of the backward-looking θ or θ_{gt} ; the assumptions below are *extrapolation* assumptions that connect those quantities to the FATT.

4.2.1 Choice of f and identifying assumptions

Path 1 extrapolation function: We specify $f_1(g) = \gamma_g$, a function that depends only on the group index g and is constant over time. The input space is simply $\mathcal{J}_t = \{g\}$; no temporal information is used. This corresponds to assuming that within-group treatment effects do not vary with calendar time or event time.

Connecting the FATT to the ATT. We now consider what assumptions would be required to identify the FATT by the ATT. First, we could require homogeneity in effects that would rule out dynamic treatment effects:

Assumption 1 (Strict ATT time homogeneity). $\mathbb{E}_{\mathbb{P}_t}\{Y_{it}(1) - Y_{it}(0)|A_{it} = 1\} = \mathbb{E}_{\mathbb{P}_s}\{Y_{is}(1) - Y_{is}(0)|A_{is} = 1\}$, for all s, t .

This assumption states that ATTs are time-invariant: the ATT at time t is the same as at time s for all s, t . We further need to place moderate limits on the between-state heterogeneity:

Assumption 2 (Limited between-state ATT heterogeneity). $\mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{ip} = 1\} = \mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{i,p+1} = 1\}$

Assumption 2 limits the between-state heterogeneity such that the treatment effect among states who had implemented the law at the end of the study period (at time p) is the same as the treatment effect among states who will have implemented the law in the future (at time $p + 1$). In most cases, these two groups of states are nearly identical: most states that implemented the law at time p keep it through to $p + 1$, and most states do not newly implement the law at time $p + 1$. So this assumption is generally relatively weak.¹

Under Assumptions 1 and 2, we have

$$\theta = \theta_{p+1}.$$

Proposition 4.1 (Identification of the FATT under time homogeneity). *Suppose the backward-looking ATT θ is identified from the observed data under the researcher's design (e.g., parallel trends in difference-in-differences). Under Assumptions 1 and 2, the FATT θ_{p+1} is identified by $\theta_{p+1} = \theta$.*

Proof. See Appendix A.2. □

Remark 4.2. *Because of the symmetry of their definitions, similar assumptions – with $A_{it} = 0$ replacing $A_{it} = 1$ and other similar replacements – would yield identification of the FATU with the ATU $\omega = \mathbb{E}_{\mathbb{P}}\{Y_{it}(1) - Y_{it}(0)|A_{it} = 0\}$.*

Remark 4.3. *Assumption 1 has testable implications. For any s, t for which $\mathbb{E}_{\mathbb{P}_s}\{Y_{is}(1) - Y_{is}(0)|A_{is} = 1\}$ and $\mathbb{E}_{\mathbb{P}_t}\{Y_{it}(1) - Y_{it}(0)|A_{it} = 1\}$ are identifiable from the observable data, it may be tested whether they are equal. If differences between different time-specific treatment effects are small, that would lend credence to Assumption 1. In contrast, Assumption 2 is untestable: it concerns $\mathbb{P}_{p+1}$, for which we have no direct information in the observed data. Identification of the FATT under this path therefore rests on an extrapolation from period p to $p + 1$ —a form of transportability across time [Pearl and Bareinboim, 2022, Egami and Hartman, 2023]—and can only be supported by substantive reasoning or indirect checks. Similar to the assessment of parallel trends in difference-in-differences [Sun and Abraham, 2021, Wang et al., 2024], we may examine whether there are any large*

¹For multi-group extrapolation (Path 2, Section 4.5), this assumption applies within each group g : units in group g who will be treated at $p + 1$ have similar potential outcomes to those treated earlier in group g . This is invoked in the proof of Proposition 4.4.

differences between $\mathbb{E}_{\mathbb{P}_{t+1}}\{Y_{i,t+1}(1) - Y_{i,t+1}(0)|A_{it} = 1\}$ and $\mathbb{E}_{\mathbb{P}_{t+1}}\{Y_{i,t+1}(1) - Y_{i,t+1}(0)|A_{i,t+1} = 1\}$ for any $t \in \{0, \dots, p-1\}$ for which these effects are identified, as indirect evidence about stability of the composition of the treated group over time.

4.3 Connecting other forward-looking estimands to backward-looking estimands

If we are willing to further restrict between-state heterogeneity, then we can find equivalence between other effects and the ATT. Consider assuming that there is no between-state heterogeneity in treatment effects at time $p+1$:

Assumption 3 (No future between-state ITE heterogeneity). $\tau_{i,p+1} = \tau_{j,p+1}$ for all i, j .

Under Assumptions 1 and 3, we have for all i, ρ, η ,

$$\theta = \omega_{p+1} = \tau_{p+1} = \tau_{i,p+1}(\eta; \rho) = \tau_{i,p+1}.$$

Under these (restrictive) assumptions, all of the proposed forward-looking estimands are equivalent to the backward-looking ATT.

Often this assumption of strict ITE homogeneity may not be believable. While this may preclude using methods for estimating the ATT to identify these future-looking estimands, it does not preclude using corresponding backward-looking versions of these estimands under weaker assumptions. For example, under assumptions analogous to Assumptions 1 and 2, one has that $\tau_{p+1} = \tau \equiv \mathbb{E}_{\mathbb{P}}\{Y_{it}(1) - Y_{it}(0)\}$ by removing the conditioning on $A_{it} = 1$ from Assumptions 1 and 2. We state this in some generality in the following assumptions.

Assumption 4 (Strict time homogeneity under $\mathcal{C}_t$). $\mathbb{E}_{\mathbb{P}_t}\{Y_{it}(1) - Y_{it}(0)|\mathcal{C}_t\} = \mathbb{E}_{\mathbb{P}_s}\{Y_{is}(1) - Y_{is}(0)|\mathcal{C}_s\}$, for some conditioning event $\mathcal{C}_t$ defined on $\mathbb{P}_t$ and all $s, t \in \{1, \dots, p+1\}$.

Assumption 5 (Limited between-state heterogeneity under $\mathcal{C}_t$). $\mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|\mathcal{C}_p\} = \mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|\mathcal{C}_{p+1}\}$

Under Assumptions 4 and 5, taking $\mathcal{C}_t = \{A_{it} = 0\}$ yields equivalence between the backward-looking ATU and the FATU; taking $\mathcal{C}_t = \emptyset$ yields equivalence between the backward-looking ATE and the FATE; and taking $\mathcal{C}_t = \{\rho(X_i, X_j) \leq \eta\}$ yields equivalence between the backward-looking ATS and the FATS (Assumption 5 not required for the FATS).

4.4 Connecting the FATT to sub-ATTs

Assumption 1 (and Assumption 4) restricts treatment effects to be constant over time. A great deal of recent literature in causal inference for panel data has explicitly rejected this assumption, positing that treatment effects may vary over time in multiple ways [Sun and Abraham, 2021, Brown and Butts, 2025]. Proposals for group-time effects θ_{gt} have attempted to capture variation in treatment effects by when the treatment was first implemented (by encoding this information in $G_i = g$) and also allowing treatment effects to vary dynamically over time thereafter.

One way to weaken Assumption 1 is to only require homogeneity within groups:

Assumption 6 (Strict within-group ATT time homogeneity). $\theta_{gt} = \theta_{gs} \equiv \theta_g$, for all $s, t \in \{1, \dots, p+1\}$.

Assumption 7 (Limited within-group between-state ATT heterogeneity). $\mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{ip} = 1, G_i = g\} = \mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{i,p+1} = 1, G_i = g\}$

Under Assumptions 6 and 7, we have that

$$\theta_{p+1} = \sum_g \mathbb{P}(G_i = g | A_{ip} = 1) \theta_g.$$

(Derivation: Appendix A.7, Lemma A.7.) We assume that G_i is a function of A_i (e.g., time of first adoption), so that $\mathbb{P}(G_i = g | A_{ip} = 1)$ is the distribution of cohorts among units treated by period p . This states that, if within-group ATTs are not dynamic – i.e., they do not change over time – then a convex combination of group-specific ATTs may be used to extrapolate forward in time to time $p+1$.

4.5 Path 2: Temporal parametric functions

Much of the recent panel data literature with staggered adoption has either explicitly [Sun and Abraham, 2021, Wang et al., 2024, Brown and Butts, 2025] or implicitly [Stock and Watson, 2018] taken the view that assuming time homogeneity is too restrictive. Acknowledgment of the perverse effects of time heterogeneity in two-way fixed effects models [Sun and Abraham, 2021, Wang et al., 2024] spurred much of the recent literature on staggered adoption in difference-in-differences. However, unrestricted time heterogeneity renders future-looking estimands like the FATT inestimable and undercuts policy relevance of any estimated quantities. Instead, the heterogeneity of treatment effects must be restricted so that information from the observed data may be used to extrapolate to the future time period [Khosrowi, 2019, 2023, Maeba, 2024]. An alternative would be to report bounds or conduct sensitivity analysis without imposing a functional form [Manski, 2013]; the parametric path we take here trades that generality for point identification and extrapolation. Researchers should prefer Path 2 over Path 1 when they are unwilling to assume that within-group effects are constant over time but are willing to specify how effects evolve (e.g., as a function of event time).

4.5.1 Choice of f and identifying assumptions

Path 2 extrapolation function: We specify $f_2(g, t; \gamma)$ as a parametric function of group g and time t (or event time $k = t - g$). The input space is $\mathcal{J}_t = \{g, t\}$, encoding both group identity and temporal dynamics. Common specifications include linear-in-event-time ($f(g, t; \gamma) = \alpha_g + \beta_g(t - g)$), quadratic, or flexible splines. The key assumption is that effects evolve according to a smooth functional form that can be extrapolated beyond the observed time window.

To estimate the FATT under Path 2, researchers must parameterize the time-varying group-time ATTs in terms of time [Brown and Butts, 2025, Maeba, 2024]. For example, one could specify

$$\theta_{gt} = f(g, t; \gamma), \quad t = 1, \dots, p+1; g \in \mathcal{G}$$

where $f(\cdot; \gamma)$ is a known function of a finite-dimensional parameter $\gamma \in \mathbb{R}^d$. The parameter γ is identified from the observed group-time ATTs θ_{gt} for $t \leq p$, $g \in \mathcal{G}$, whenever the mapping from γ to $(\theta_{gt})_{g,t}$ (restricted to $t \leq p$) is injective, so that γ is uniquely determined by fitting the model to the identified θ_{gt} . A common choice is to model effects as a function of event time $k = t - g$, e.g., $\theta_{gt} = h_g(k; \gamma)$ with h_g linear or quadratic in k ; then $f(g, t; \gamma) = h_g(t - g; \gamma)$. Then the FATT may be identified as

$$\theta_{p+1} = \sum_g \mathbb{P}(G_i = g | A_{ip} = 1) f(g, p+1; \gamma).$$

Proposition 4.4 (Identification of the FATT under parametric dynamics). *Suppose the group-time ATTs θ_{gt} are identified from the observed data under the researcher’s design for $t \leq p$, $g \in \mathcal{G}$. Suppose $\theta_{gt} = f(g, t; \gamma)$ with γ uniquely determined by $(\theta_{gt})_{g,t,t \leq p}$. Then the FATT θ_{p+1} is identified by $\theta_{p+1} = \sum_g \mathbb{P}(G_i = g | A_{ip} = 1) f(g, p+1; \gamma)$.*

Proof. See Appendix A.3. □

4.6 Path 3: Covariate-distribution functions

Both Path 1 (time homogeneity) and Path 2 (parametric extrapolation of temporal patterns) rely on reduced-form stability: either effects are constant over time, or they follow a smooth trend that continues into the future. These assumptions fail under *regime change*—structural shifts in the policy environment, implementation context, or economic conditions that break historical patterns. The Lucas Critique [Lucas Jr, 1976] emphasizes that reduced-form relationships estimated under one policy regime need not hold under another. If the data-generating process itself changes at $p+1$, extrapolating reduced-form time trends (Path 2) or assuming temporal stability (Path 1) may yield biased estimates of the FATT.

Path 3 offers an alternative: instead of assuming temporal invariance of effects, we assume that treatment effects are driven by observable covariates that capture “deep parameters”—structural determinants that remain invariant even when the policy regime changes. If heterogeneity in treatment effects is well-explained by pre-treatment covariates X_i , and if the conditional treatment effect $\tau(x) = \mathbb{E}[Y_{i,p+1}(1) - Y_{i,p+1}(0) | X_i = x]$ is stable across periods (including at $p+1$), then the FATT can be identified by estimating $\tau(x)$ from historical data and integrating it over the *future* covariate distribution. This approach is robust to regime change in the sense that temporal patterns in the marginal ATTs θ_{gt} may shift (violating Paths 1 and 2), but the conditional structure $\tau(x)$ remains predictive. This is the essence of the Lucas Critique’s prescription: estimate parameters that are invariant to the policy intervention itself.

4.6.1 Choice of f and identifying assumptions

Path 3 extrapolation function: We specify $f_3(F_X^t; \beta) = \int m(x; \beta) dF_X^t(x)$, where F_X^t is the covariate distribution at time t . The input space is $\mathcal{X}_t = \{F_X^t\}$, the population composition at time t . The key structural object is the conditional effect function $m(x; \beta)$, which we assume is invariant across time. Temporal variation in the marginal effect θ_{gt} arises solely through changes in the covariate distribution F_X^t , not through changes in the conditional effect itself. This is the deep-parameter assumption of the Lucas Critique: $m(x; \beta)$ captures regime-invariant relationships, while F_X^t may shift with policy or economic changes.

Identification via covariate integration. Let X_i denote pre-treatment covariates (either time-invariant or, if time-varying, forecasted to $p + 1$). Define the conditional ATT at the future period:

$$\tau(x) = \mathbb{E}_{\mathbb{P}_{p+1}}[Y_{i,p+1}(1) - Y_{i,p+1}(0) | X_i = x, A_{i,p+1} = 1].$$

The FATT for units with treatment status $A_{ip} = 1$ (i.e., units treated by the end of the study period, who continue treatment at $p + 1$) is then

$$\theta_{p+1} = \int \tau(x) dF_{X|A_{ip}=1}^{p+1}(x), \quad (1)$$

where $F_{X|A_{ip}=1}^{p+1}$ is the conditional distribution of X among units with $A_{ip} = 1$ at time $p + 1$. If covariates are time-invariant, this is simply the empirical distribution of X_i among treated units in the study sample. If covariates are time-varying or the target population differs (e.g., we wish to extrapolate to a different jurisdiction), F_X^{p+1} may differ from the historical distribution and must be specified or forecasted.

The key identifying assumptions are as follows.

Assumption 8 (Structural stability of conditional effects). *The conditional ATT function $\tau(x)$ is invariant across periods $t \in \{1, \dots, p, p + 1\}$. That is, for all x in the support of X and all t ,*

$$\mathbb{E}_{\mathbb{P}_t}[Y_{it}(1) - Y_{it}(0) | X_i = x, A_{it} = 1] = \tau(x).$$

Assumption 8 is the analog of time homogeneity (Assumption 1) but is conditional on covariates rather than marginal over time. It states that the effect of treatment for units with covariate value x does not change over time, even though the marginal distribution of X (and hence the marginal ATT θ_t) may shift. This is weaker than marginal time homogeneity when covariate-driven selection or composition changes occur: for instance, if early adopters differ systematically from later adopters in observed X , Assumption 8 can hold even when marginal θ_t varies over time. Conversely, if unobserved factors drive time-varying effects that are not captured by X , this assumption fails.

Importantly, Assumption 8 implies *conditional exchangeability*: conditional on covariates $X = x$, potential outcomes at $p + 1$ are governed by the same function $\tau(x)$ as in observed periods. This means that among units with $X_i = x$, those who will be treated at $p + 1$ and those treated in earlier periods have similar potential outcomes—analogue to the limited between-state heterogeneity assumption (Assumption 2), but applied conditionally on x rather than marginally. This conditional exchangeability is invoked in the proof of Proposition 4.5.

Assumption 9 (Parametric specification of $\tau(x)$). *The conditional ATT has a known functional form $\tau(x) = m(x; \beta)$, where m is a specified function and $\beta \in \mathbb{R}^d$ is a finite-dimensional parameter.*

Assumption 9 is analogous to the parametric model for temporal dynamics in Path 2 (Assumption 3, where we specified $\theta_{gt} = f(g, t; \gamma)$). Common choices for m include linear-in-covariates models, $\tau(x) = x^\top \beta$, or additive models with basis expansions. The parameter β must be identified from observed data; we return to this point below.

Assumption 10 (Access to target covariate distribution). *Either (a) we observe a finite sample $\{X_i^*\}_{i=1}^{n^*}$ drawn from F_X^{p+1} , or (b) we can generate draws from F_X^{p+1} via forecasting or simulation.*

Assumption 10 ensures that the integral in (1) is computable. In many applications, covariates are time-invariant (e.g., geographic or demographic characteristics of units), so the target distribution is simply the empirical distribution of X_i among treated units in the sample. In other settings, covariates may be forecasted (e.g., population growth, economic indicators), or we may wish to transport the estimated $\tau(x)$ to a different target population with a known or estimable covariate distribution.

We now state the identification result.

Proposition 4.5 (Identification via covariate integration). *Suppose the group-time ATTs θ_{gt} are identified from the observed data under the researcher's design for $t \leq p$, $g \in \mathcal{G}$. Suppose Assumptions 8–10 hold, and suppose further that the mapping $\beta \mapsto (\theta_{g1}, \dots, \theta_{gp})_{g \in \mathcal{G}}$ is injective, so that β is uniquely determined by the observed group-time ATTs via the relationship*

$$\theta_{gt} = \int m(x; \beta) dF_{X|G_i=g}(x), \quad t \in \{g, \dots, p\}, g \in \mathcal{G}.$$

Then the FATT θ_{p+1} is identified by

$$\theta_{p+1} = \int m(x; \beta_0) dF_{X|A_{ip}=1}^{p+1}(x),$$

where β_0 is the true parameter satisfying the above system of equations.

Proof. See Appendix A.4. □

Interpretation and credibility. Path 3 shifts the extrapolation burden from time to covariates. Instead of assuming that temporal patterns continue (Path 2) or that effects are stable over time (Path 1), we assume that the conditional effect $\tau(x)$

is regime-invariant. This is credible when X captures the key drivers of treatment effect heterogeneity—what the Lucas Critique calls “deep parameters.” For example, if the effect of a policy depends on units’ economic conditions, demographic composition, or institutional capacity, and these are observed in X , then $\tau(x)$ may be stable even when the policy regime changes. On the other hand, if treatment effects are driven by unobserved time-varying factors (e.g., political sentiment, enforcement intensity, or spillover effects not captured by X), Path 3 will fail just as Paths 1 and 2 do.

The trade-off is as follows: Path 1 requires marginal temporal stability (strong when composition shifts); Path 2 requires extrapolating a smooth temporal trend (strong when regime change breaks trends); Path 3 requires correct specification of the covariate-driven heterogeneity (strong when key drivers are unobserved or when the functional form $m(x; \beta)$ is misspecified). Researchers should choose the path that aligns with what they believe is invariant in their setting. Path 3 is particularly compelling in contexts where: (i) substantial covariate-driven heterogeneity is documented; (ii) the policy environment is shifting (so Paths 1 and 2 are dubious); (iii) the key sources of heterogeneity are observable and well-measured.

4.7 Semiparametric estimation and EIF

Under the parametric model $m(x; \beta)$, we estimate β by fitting the model to the observed group-time ATTs θ_{gt} and their covariate distributions within groups. For instance, if we specify $\tau(x) = x^\top \beta$ (linear-in-covariates), then for each cohort g and period $t \geq g$, we have

$$\theta_{gt} = \int x^\top \beta dF_{X|G_i=g}(x) = \mathbb{E}[X_i | G_i = g]^\top \beta.$$

This yields a system of equations: $\theta_{gt} = \bar{X}_g^\top \beta$ for each (g, t) , where $\bar{X}_g = \mathbb{E}[X_i | G_i = g]$. If the design matrix formed by stacking $\bar{X}_g$ (or more generally, $\bar{X}_{gt}$ if covariates vary within group-time cells) has full column rank, β is identified. More generally, for nonlinear $m(x; \beta)$, we fit β via weighted least squares or method-of-moments: minimize

$$\sum_{g,t} w_{gt} \left(\hat{\theta}_{gt} - \int m(x; \beta) d\hat{F}_{X|G_i=g}(x) \right)^2,$$

where w_{gt} are weights (e.g., inverse variance or sample size of group-time cells) and $\hat{F}_{X|G_i=g}$ is the empirical covariate distribution within group g .

Once $\hat{\beta}$ is obtained, the FATT estimator is

$$\hat{\theta}_{p+1} = \begin{cases} \frac{1}{n^*} \sum_{i=1}^{n^*} m(X_i^*; \hat{\beta}) & \text{(finite-population target sample)} \\ \frac{1}{M} \sum_{j=1}^M m(X_j; \hat{\beta}) & \text{(Monte Carlo integration),} \end{cases} \quad (2)$$

where $\{X_i^*\}_{i=1}^{n^*}$ is the target sample (e.g., treated units in the study, or a new target population), or $\{X_j\}_{j=1}^M$ are draws from a

sampler for F_X^{p+1} .

Path 3 EIF The efficient influence function for Path 3 is derived via the chain rule, accounting for estimation of β from the group-time ATTs and integration over the target covariate distribution. By the functional delta method (see Appendix A.5.3), the EIF for $\psi_3 = \theta_{p+1}$ is

$$\phi_{\psi_3} = \left(\frac{\partial \Psi_3}{\partial \beta} \right)^\top \left(\sum_{g,t} \frac{\partial \beta}{\partial \theta_{gt}} \phi_{gt} \right) + \sum_{g,t} \left(\frac{\partial \Psi_3}{\partial \omega_{gt}} \right)^\top \phi_{\omega_{gt}},$$

where $\Psi_3(\theta, \omega, \beta) = \int m(x; \beta) dF_X^{p+1}(x)$ and the Jacobian terms capture: (i) the derivative of the FATT with respect to β ; (ii) the implicit derivative of β with respect to each θ_{gt} (via the method-of-moments or least-squares fitting); (iii) if covariate distributions or weights are estimated, their influence functions. For the finite-population case in (2), we have

$$\frac{\partial \Psi_3}{\partial \beta} = \frac{1}{n^*} \sum_{i=1}^{n^*} \frac{\partial m(X_i^*; \beta)}{\partial \beta}.$$

For the linear-in-covariates case $m(x; \beta) = x^\top \beta$, this simplifies to $\bar{X}^*$, the mean of the target covariates. The implicit function theorem gives $\frac{\partial \beta}{\partial \theta_{gt}}$ by differentiating the first-order conditions for the least-squares or method-of-moments fit; the resulting Jacobian is a function of the Hessian of the objective and the gradient with respect to θ_{gt} . Full details are in Appendix A.5.3.

Within the parametric model $m(x; \beta)$ and conditional on the target covariate distribution F_X^{p+1} , the EIF ϕ_{ψ_3} achieves the semiparametric variance bound. If m is misspecified, the EIF is valid for the pseudo-true parameter (the limit of $\hat{\beta}$ under the misspecified model), but $\hat{\theta}_{p+1}$ need not be consistent for the true FATT. Researchers should assess model fit and consider sensitivity to the choice of m (e.g., linear vs nonlinear, which covariates to include).

When to use Path 3. Path 3 is most compelling when: (i) substantial treatment effect heterogeneity is driven by observable covariates; (ii) temporal stability (Path 1) is implausible due to composition changes or regime shifts; (iii) extrapolating temporal trends (Path 2) is dubious because the policy environment is changing; (iv) the key drivers of heterogeneity are observed and well-measured, so that $\tau(x)$ is likely regime-invariant. Path 3 is less appealing when heterogeneity is driven by unobserved factors, when the functional form $m(x; \beta)$ is uncertain, or when the target covariate distribution F_X^{p+1} is poorly known.

5 Semiparametric Estimation and Inference

All three identification paths require estimation and inference for the FATT θ_{p+1} . We do not commit to a specific identification or first-stage estimation strategy. We assume that the group-time ATTs θ_{gt} are identified under the researcher's chosen design

(e.g., difference-in-differences, synthetic control, or other methods); we do not restrict the identification strategy. We further assume that for each (g, t) , the researcher has access to an estimator $\hat{\theta}_{gt}$ that is asymptotically linear with influence function ϕ_{gt} : $\sqrt{n}(\hat{\theta}_{gt} - \theta_{gt}) = n^{-1/2} \sum_{i=1}^n \phi_{gt,i} + o_{\mathbb{P}}(1)$. Such estimators exist in the literature for various designs [Callaway and Sant’Anna, 2021] and may be constructed with or without cross-fitting; our results hold for any such first-stage estimators.

5.1 Efficient influence functions

The contribution of this subsection is to derive the efficient influence function (EIF) for the policy-relevant FATT θ_{p+1} as a functional of the group-time ATTs and their influence functions, so that any asymptotically linear estimators of θ_{gt} yield an asymptotically linear estimator of θ_{p+1} with limiting variance given by the propagated EIF. We present the EIF for Paths 1 and 2 here; the EIF for Path 3 (covariate integration) was derived in Section 4.6.

Path 1 (time homogeneity). Under within-group time homogeneity, $\theta_{p+1} = \sum_g \omega_g \theta_g$, with $\omega_g = \mathbb{P}(G_i = g | A_{ip} = 1)$. Take as given the EIF ϕ_{θ_g} for each θ_g . (e.g., averaged over t in the observed window, or as provided by the first-stage). By linearity (see Appendix A.5.1), the EIF for $\psi_1 = \theta_{p+1}$ is

$$\phi_{\psi_1} = \sum_g \omega_g \phi_{\theta_g} + \sum_g \theta_g \phi_{\omega_g},$$

where ϕ_{ω_g} is the influence function of the estimator of ω_g (e.g., for the empirical proportion among units with $A_{ip} = 1$, unit i contributes $1\{G_i = g, A_{ip} = 1\} / \mathbb{P}(A_{ip} = 1) - \omega_g$); if ω_g is known, the second sum vanishes. Thus any asymptotically linear estimators of θ_g yield an asymptotically linear estimator of ψ_1 with the stated influence function.

Path 2 (extrapolation). Under $\theta_{gt} = f(g, t; \gamma)$, we have $\theta_{p+1} = \Psi(\theta, \omega) = \sum_g \omega_g f(g, p+1; \gamma)$, where γ is a smooth functional of $\theta = (\theta_{g,t})_{g,t}$. Take as given the EIF $\phi_{\theta_{gt}}$ for each θ_{gt} from whatever first-stage the researcher uses. By the chain rule (see Appendix A.5.2), the EIF for $\psi_2 = \theta_{p+1}$ is

$$\phi_{\psi_2} = \frac{\partial \Psi}{\partial \theta} \phi_{\theta} + \frac{\partial \Psi}{\partial \omega} \phi_{\omega},$$

with the obvious vectorization over (g, t) and, if ω_g is estimated, an additional term $\sum_g f(g, p+1; \gamma) \phi_{\omega_g}$. The Jacobian $\frac{\partial \Psi}{\partial \theta}$ is the row vector with entries $\frac{\partial \Psi}{\partial \theta_{gt}} = \frac{\partial \Psi}{\partial \gamma}^\top \frac{\partial \gamma}{\partial \theta_{gt}}$, where $\frac{\partial \Psi}{\partial \gamma} = \sum_g \omega_g \frac{\partial f(g, p+1; \gamma)}{\partial \gamma}$; thus stacking the $\phi_{\theta_{gt}}$ and multiplying by this row gives the contribution from the first-stage. Within the model that restricts $\theta_{gt} = f(g, t; \gamma)$, this ϕ is the efficient influence function for θ_{p+1} .

Proposition 5.1 (Asymptotic distribution of $\hat{\theta}_{p+1}$). *Under the assumptions of Path 1, Path 2, or Path 3 above, suppose the first-stage estimators $\hat{\theta}_{gt}$ (and, where relevant, $\hat{\omega}_g$, $\hat{\gamma}$, or $\hat{\beta}$) are asymptotically linear with the influence functions stated. Then*

the estimator $\widehat{\psi} = \widehat{\theta}_{p+1}$ constructed by plugging these into the corresponding ϕ_{ψ_1} , ϕ_{ψ_2} , or ϕ_{ψ_3} satisfies $\sqrt{n}(\widehat{\psi} - \psi) \rightsquigarrow N(0, \sigma^2)$ with $\sigma^2 = \mathbb{E}[\phi^2]$.

Proof. See Appendix A.6. □

Variance and asymptotics. The semiparametric variance bound is $\sigma^2 = \mathbb{E}[\phi^2]$. Under standard regularity conditions (see Appendix A.1), $\sqrt{n}(\widehat{\psi} - \psi) = n^{-1/2} \sum_{i=1}^n \phi_i + o_{\mathbb{P}}(1)$, so $\sqrt{n}(\widehat{\psi} - \psi) \rightsquigarrow N(0, \sigma^2)$. The variance may be estimated by $\widehat{\sigma}^2/n$ with $\widehat{\sigma}^2 = n^{-1} \sum_i \widehat{\phi}_i^2$ (or the sample variance of $\widehat{\phi}_i$); Wald confidence intervals are $\widehat{\psi} \pm z_{\alpha/2} \widehat{\sigma} / \sqrt{n}$. The estimated $\widehat{\phi}_i$ is a plug-in that uses $\widehat{\theta}_{gt}$, $\widehat{\gamma}$, and possibly $\widehat{\omega}_g$; under standard regularity, this does not affect the first-order limit distribution. If the researcher uses flexible (e.g., machine-learning) estimation when forming $\widehat{\theta}_{p+1}$ (e.g., when estimating γ or ω_g), K -fold cross-fitting can relax Donsker requirements: the aggregation or extrapolation step (estimate γ or ω_g on folds $\mathcal{J}_{-k}$, evaluate the EIF on $\mathcal{J}_k$) is cross-fit; only rate conditions (e.g., $o_{\mathbb{P}}(n^{-1/4})$) are needed. The paper does not require cross-fitting for the first-stage $\widehat{\theta}_{gt}$; the results hold for any asymptotically linear first-stage. A limitation of the parametric extrapolation approach is that bias and inference can be sensitive to the choice of f : if the temporal model is misspecified, $\widehat{\theta}_{p+1}$ need not be consistent for the FATT.

5.2 Model selection for extrapolation

Path 2 requires the researcher to specify a parametric form $f(g, t; \gamma)$ for the temporal evolution of group-time effects. The choice of f is substantive: it encodes the researcher’s belief about how treatment effects evolve over event time or calendar time. Common specifications include linear-in-event-time, quadratic, or flexible splines; the extrapolated FATT $\widehat{\theta}_{p+1}$ can differ substantially across these choices, especially when projecting multiple periods ahead. This section addresses a practical question: *how should researchers select among candidate models for extrapolation?*

The problem with standard model selection criteria. Standard approaches such as AIC, BIC, or cross-validated in-sample prediction error measure fit *within* the observed data. For model selection aimed at extrapolation, these criteria are misaligned with the task: a model that fits the observed θ_{gt} for $t \leq p$ well may extrapolate poorly to $t = p + 1$. Overfitting to the observed time pattern can produce overconfident extrapolations that fail when the true functional form differs from f . Conversely, underfitting (e.g., imposing strict linearity when effects are nonlinear) can introduce systematic bias in the extrapolated estimate. What is needed is a criterion that directly evaluates *extrapolation performance*.

Time-series cross-validation for extrapolation. We propose a time-series cross-validation (CV) strategy that leverages the temporal structure of the observed data to test extrapolation. The key idea is to withhold late-period observations, fit candidate models on earlier periods, and evaluate prediction error on the held-out future periods—thereby mimicking the extrapolation task.

Let $\mathcal{F} = \{f_1, f_2, \dots, f_M\}$ denote a set of candidate models (e.g., f_1 is linear-in-event-time, f_2 is quadratic, f_3 is a spline). For each model f_m , define the extrapolation error on held-out periods as follows:

1. **Training window:** Fit the model using θ_{gt} for $t \in \{1, \dots, p-h\}$, estimating parameters $\hat{\gamma}_m^{(h)}$ via weighted least squares or maximum likelihood.
2. **Test window:** Predict $\hat{\theta}_{gt}^{(h)} = f_m(g, t; \hat{\gamma}_m^{(h)})$ for the held-out periods $t \in \{p-h+1, \dots, p\}$.
3. **Extrapolation error:** Compute the mean squared prediction error (MSPE) on the test window:

$$\text{MSPE}_m(h) = \frac{1}{|\mathcal{G}| \cdot h} \sum_{g \in \mathcal{G}} \sum_{t=p-h+1}^p \left(\hat{\theta}_{gt}^{(h)} - \theta_{gt} \right)^2,$$

where θ_{gt} are the observed (first-stage estimated) group-time effects.

4. **Aggregation:** Repeat for multiple horizons $h \in \{1, 2, \dots, H\}$ (e.g., holding out the last 1, 2, or 3 periods). The overall extrapolation error for model m is

$$\text{MSPE}_m = \frac{1}{H} \sum_{h=1}^H \text{MSPE}_m(h).$$

Model selection rule: Select the model $\hat{m} = \arg \min_{m \in \{1, \dots, M\}} \text{MSPE}_m$. This model minimizes the average extrapolation error across the validation experiments.

Coverage-based model selection. An alternative or complementary criterion is to evaluate whether the model's confidence intervals achieve nominal coverage on held-out periods. For each candidate model f_m and horizon h , construct prediction intervals for θ_{gt} in the test window (e.g., using the EIF-based variance from Section 5.1). Define the *coverage rate* as the proportion of held-out (g, t) pairs for which the 95% confidence interval contains the observed θ_{gt} :

$$\text{Coverage}_m(h) = \frac{1}{|\mathcal{G}| \cdot h} \sum_{g \in \mathcal{G}} \sum_{t=p-h+1}^p \mathbb{1} \left\{ \theta_{gt} \in \left[\hat{\theta}_{gt}^{(h)} \pm 1.96 \cdot \widehat{\text{SE}}_{gt}^{(h)} \right] \right\}.$$

A model with high extrapolation MSPE but poor coverage may be overfitting and underestimating uncertainty; a model with coverage near 95% and low MSPE is preferable.

Practical workflow. We recommend the following approach for researchers using Path 2:

1. **Specify candidate models:** Choose a small set of plausible functional forms based on substantive knowledge (e.g., linear, quadratic, log-linear in event time; or splines with varying degrees of freedom).
2. **Run time-series CV:** Apply the procedure above to compute MSPE_m and Coverage_m for each candidate model. If the time series is short (e.g., $p \leq 5$), hold out only 1–2 periods; if longer, hold out more to test longer-horizon extrapolation.

3. **Select or average:** Either select the single best model $\widehat{m}$ (lowest MSPE, coverage near nominal), or report a model-averaged estimate $\widehat{\theta}_{p+1} = \sum_m w_m \widehat{\theta}_{p+1,m}$ where $w_m \propto \exp(-\text{MSPE}_m)$ (exponential weighting). Model averaging acknowledges model uncertainty and can improve robustness.
4. **Report sensitivity:** In addition to the selected or averaged estimate, report $\widehat{\theta}_{p+1}$ under the top 2–3 models to convey sensitivity to functional form. If estimates differ substantially, the extrapolation is highly model-dependent and should be interpreted with caution.
5. **Graphical diagnostics:** Plot the fitted $f(g, t; \widehat{\gamma})$ against observed θ_{gt} for each group g , extending the curves to $t = p + 1$. Visual inspection can reveal whether the extrapolation is a modest continuation of observed trends or a large leap beyond the data.

Limitations and caveats. Time-series CV evaluates how well a model extrapolates *within the observed time window*. It does not, and cannot, test extrapolation to $t = p + 1$ if that period involves regime change, distributional shifts, or other violations of the structural stability assumptions underlying Path 2. The CV procedure assumes that the data-generating process for θ_{gt} is stable across $t \leq p$, so that performance on held-out $t \in \{p - h + 1, \dots, p\}$ is predictive of performance at $t = p + 1$. If $\mathbb{P}_{p+1}$ differs markedly from $\mathbb{P}_t$ for $t \leq p$ —for example, due to policy reforms, economic shocks, or compositional changes in the treated population—then even the best model selected via CV may be biased.

Moreover, if the true functional form is not in the candidate set $\mathcal{F}$, CV selects the best approximation within $\mathcal{F}$, but this may still yield biased extrapolations. Researchers should use substantive knowledge to ensure that $\mathcal{F}$ includes plausible models and, where possible, include flexible forms (e.g., splines) that can approximate a wide class of smooth functions.

Finally, the MSPE criterion averages over groups and time; it may mask heterogeneity in extrapolation performance across groups. If different groups exhibit different temporal patterns (e.g., one group has linear effects, another quadratic), a single global model $f(g, t; \gamma)$ may extrapolate poorly for some groups even if it minimizes average MSPE. Group-specific diagnostics (plotting fitted trends for each g) can help identify such cases.

Relation to post-selection inference. The CV procedure selects a model based on the data, introducing an additional source of randomness beyond the first-stage estimation of θ_{gt} . Formally accounting for model selection in inference (e.g., via selective inference frameworks or simultaneous inference over $\mathcal{F}$) is an active area of research [??]. The influence function derivation in Section 5.1 conditions on the chosen model f and thus does not account for the uncertainty introduced by selection. In practice, if the selected model is stable across CV folds and clearly dominates alternatives, the impact of selection on coverage may be modest. If selection is highly variable or multiple models perform similarly, model averaging or simultaneous confidence bands may be warranted. We leave formal post-selection inference for Path 2 to future work, but emphasize that the CV-based workflow provides a principled empirical foundation for model choice that is aligned with the extrapolation task.

6 Simulations

Policy-relevant questions concern the FATT at $p + 1$, whereas standard panel-data methods target backward-looking objects such as the ATT or group-time effects over the observed window. We use simulations to show when and how the three paths—time homogeneity (Path 1), parametric extrapolation (Path 2), and covariate integration (Path 3)—succeed or fail to identify the FATT, and that EIF-based inference is valid when the model is correct. All designs use $q = 3$ groups, $p = 5$ periods, $n = 500$ units, and 1000 replicates, except where noted.

6.1 Data generation

We consider q cohorts (groups) and calendar times $1, \dots, p$. For each group g , we observe group-time effects θ_{gt} for $t \geq g$ (post-treatment only), with event time $k = t - g$. The true effects θ_{gt} are generated from one of three specifications in event time: (i) constant within group, $\theta_{gt} = \theta_g$; (ii) linear, $\theta_{gt} = \alpha_g + \beta_g k$; (iii) quadratic, $\theta_{gt} = \alpha_g + \beta_g k + \delta_g k^2$. The strength of dynamics is thus controlled by the slope (and curvature) in event time. We take the role of a first-stage that has produced estimates $\hat{\theta}_{gt}$ and their influence functions. For simplicity we set $\hat{\theta}_{gt} = \theta_{gt} + \varepsilon_{gt}$ with $\varepsilon_{gt} \sim N(0, \sigma_\tau^2)$ and construct EIF vectors so that the variance of each cell estimator matches σ_τ^2 ; aggregation and extrapolation then use the EIF propagation in Sections 5–6. The simulations therefore focus on the aggregation and extrapolation step rather than on a specific difference-in-differences or other first-stage. The true FATT at $p + 1$ is $\theta_{p+1} = \sum_g \omega_g \theta_{g,p+1}$; we use equal group weights $\omega_g = 1/q$. The true backward-looking ATT is the (equal-weight) average of θ_{gt} over observed (g, t) .

The first simulation illustrates that the *target* of estimation matters: the same data can support an unbiased estimate of the backward-looking ATT while that number diverges from the FATT when effects evolve with event time. When the slope in event time is non-zero, the two curves in Figure 1 separate; interpreting the backward-looking ATT as the policy-relevant effect is then misleading. This motivates the need for time-homogeneity assumptions (Path 1), explicit temporal extrapolation (Path 2), or covariate-driven effects (Path 3).

Path 1 assumes within-group time homogeneity and forms $\hat{\theta}_{p+1} = \sum_g \omega_g \bar{\theta}_{g\cdot}$, where $\bar{\theta}_{g\cdot}$ is the average of $\hat{\theta}_{gt}$ over t for group g . Scenario A (constant effects within group) shows small bias and coverage near 95%, consistent with Section 4.2.

Table 2: Path 1: time homogeneity (Scenario A) vs dynamics (Scenario B).

Scenario	True FATT	Bias	RMSE	95% coverage
A (homogeneity)	0.133	0.001	0.031	94.10 %
B (dynamics)	0.537	-0.313	0.315	0.00 %

Scenario B (linear dynamics) shows substantial bias and coverage collapsing to zero for the FATT—Path 1 then estimates a different object (the constant-within-group extrapolant) and is invalid for the FATT. This underscores that Path 1 is only appropriate when time homogeneity is plausible.

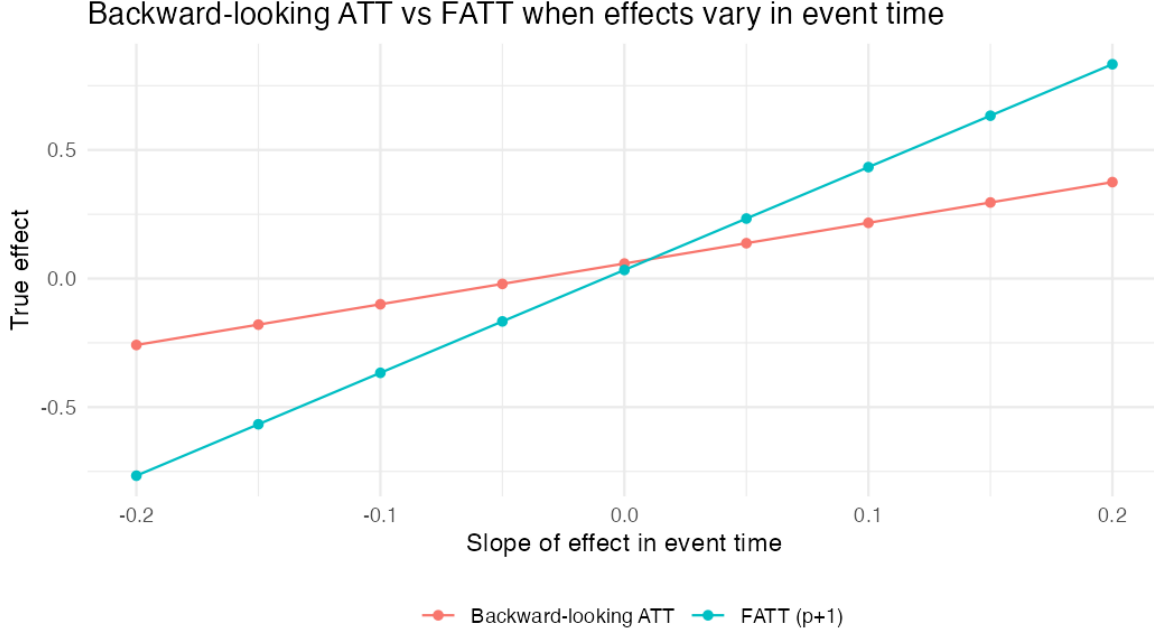


Figure 1: True backward-looking ATT and true FATT at $p + 1$ as a function of the slope of the effect in event time (linear DGP). When the slope is non-zero, the two estimands differ; interpreting the backward-looking ATT as the policy-relevant effect is misleading.

Path 2 fits a parametric model $f(g, t; \hat{\gamma})$ to $\hat{\theta}_{g,t}$ and extrapolates to $p + 1$. Correct specification (linear DGP, linear fit)

Table 3: Path 2: correct specification (linear DGP, linear fit), misspecified (quadratic DGP, linear fit), and quadratic fit (quadratic DGP, quadratic fit).

Specification	True FATT	Bias	RMSE	95% coverage
Correct (linear fit)	0.427	-0.001	0.074	95.70 %
Misspecified (linear on quad)	0.710	-0.147	0.166	49.70 %
Quadratic fit on quad	0.710	0.008	0.189	94.30 %

yields low bias and nominal coverage. Misspecification (quadratic DGP, linear fit) introduces bias and can distort coverage; fitting the correct quadratic model restores performance. This illustrates that Path 2 trades robustness for point identification and that the choice of f matters in practice.

Under correct specification, the EIF-based variance and Wald intervals from Sections 5–6 should achieve nominal coverage. The variance ratio near one and coverage near 95% and 90% confirm that the propagated EIF delivers valid inference when the first-stage and (for Path 2) the extrapolation model are correct.

On a single DGP with mild dynamics, we compare both paths. Path 2 reduces bias for the FATT relative to Path 1 when effects evolve with event time; Path 1’s simplicity is attractive when dynamics are absent, but when they are present, extrapolation can improve alignment with the policy-relevant FATT.

Finally, we demonstrate Path 3’s advantage under regime change (Table 6). We construct a DGP where effects are driven

Table 4: EIF-based variance and Wald coverage (Path 2, correct specification).

Metric	Value
Variance ratio (est./emp.)	0.95
95% coverage	94.90 %
90% coverage	89.00 %

Table 5: Path 1 vs Path 2 on the same DGP with mild dynamics (target: true FATT).

Estimator	Bias	RMSE	95% coverage
Path 1	-0.149	0.152	0.00 %
Path 2	0.003	0.071	95.40 %

by a structural covariate: $\tau(X) = 2 + 1.5 \cdot X$. Groups differ in their covariate distributions ($\mu_g \in \{-1, 0, 1\}$), so group-time ATTs vary by composition, not true time dynamics. At $p + 1$, the target distribution shifts to $\mu_{\text{target}} = 0.5$ (mimicking regime change or policy targeting high- X units). The true FATT under the new distribution is 2.75, while the historical backward-looking ATT is 1.75—a regime-change gap of 1.0. Paths 1 and 2 are heavily biased (bias ≈ -0.75 , coverage 0%) because

Table 6: Path 3 under regime change: covariate-driven effects when target distribution shifts. True FATT = 2.750 (target $\mu = 0.500$); backward-looking ATT = 1.750 (historical $\bar{\mu} \approx 0$). Regime change gap = 1.000. Paths 1 and 2 fail (biased, zero coverage); Path 3 succeeds ($\tau(X)$ is regime-invariant).

Path	True FATT	Bias	RMSE	95% coverage
1: Time homogeneity	2.750	-0.751	0.751	0.00 %
2: Temporal extrapolation	2.750	-0.753	0.757	0.00 %
3: Covariate integration	2.750	0.046	0.060	99.20 %

they rely on temporal patterns that break under regime change. Path 3 is nearly unbiased and achieves reasonable coverage (oracle version) because $\tau(X)$ is structurally invariant—it captures the deep parameters that remain predictive even when the covariate distribution shifts. This scenario illustrates the Lucas Critique: reduced-form temporal relationships fail under regime change, but structural covariate-driven effects can succeed if the key drivers of heterogeneity are observed and regime-invariant.

Taken together, the simulations reinforce the paper’s message: the target of estimation (FATT vs backward-looking ATT) matters; Paths 1, 2, and 3 each have their place depending on what the researcher believes is invariant (temporal stability, smooth trends, or structural covariates); and EIF-based inference is well-calibrated under correct specification.

6.2 Model selection via cross-validation

This section demonstrates the time-series cross-validation procedure proposed in Section 5.2. We simulate data where the true group-time effects follow a known functional form, apply the CV procedure to select among candidate models, and evaluate whether CV identifies the correct model.

Data-generating process. We generate group-time ATTs θ_{gt} for $G = 3$ groups observed over $p = 10$ periods. The true functional form is *quadratic in event time*:

$$\theta_{gt} = \alpha_g + \beta_g(t - g) + \gamma_g(t - g)^2,$$

where $\alpha_g \in \{-0.5, 0, 0.5\}$ are baseline effects, $\beta_g \in \{0.3, 0.4, 0.2\}$ are linear slopes, and $\gamma_g \in \{-0.05, -0.03, -0.04\}$ are quadratic decay rates (negative values indicate effects that peak and then decline). First-stage estimates $\hat{\theta}_{gt}$ are simulated by adding independent Gaussian noise with standard error 0.1 to the true θ_{gt} .

Candidate models. We consider three models:

1. **Linear:** $f_1(g, t; \gamma) = \alpha_g + \beta_g(t - g)$ (underspecified).
2. **Quadratic:** $f_2(g, t; \gamma) = \alpha_g + \beta_g(t - g) + \gamma_g(t - g)^2$ (correctly specified).
3. **Spline:** $f_3(g, t; \gamma)$ uses natural splines with 4 degrees of freedom (flexible).

All models are estimated via weighted least squares with weights $1/\widehat{\text{SE}}_{gt}^2$.

Cross-validation procedure. For each model $m \in \{1, 2, 3\}$ and horizon $h \in \{1, 2, 3\}$:

1. Fit the model on periods $t \leq p - h$.
2. Predict $\hat{\theta}_{gt}^{(h)}$ for test periods $t \in \{p - h + 1, \dots, p\}$.
3. Compute mean squared prediction error: $\text{MSPE}_m(h) = \frac{1}{|g| \cdot h} \sum_{g,t} (\hat{\theta}_{gt}^{(h)} - \theta_{gt})^2$.

The average MSPE is $\text{MSPE}_m = \frac{1}{3} \sum_{h=1}^3 \text{MSPE}_m(h)$. The model with the lowest average MSPE is selected.

Results. Table 7 shows the MSPE for each model and horizon. The quadratic model achieves the lowest average MSPE (0.456), substantially outperforming the linear model (0.841) and the spline (0.597). Cross-validation correctly identifies the quadratic model as the best fit. Extrapolating to period $p + 1$ using the selected quadratic model yields $\hat{\theta}_{p+1} = -0.573$, close to the true value $\theta_{p+1} = -0.596$ (error: 0.023).

Table 7: Model selection via time-series cross-validation: True DGP is quadratic

Model	$h = 1$	$h = 2$	$h = 3$	Avg MSPE	Selected
Linear	0.800	0.814	0.909	0.841	✓
Quadratic	0.431	0.490	0.448	0.456	
Spline (df=4)	0.466	0.722	0.603	0.597	

Robustness: True model not in candidate set. We repeat the exercise with a *cubic* DGP: $\theta_{gt} = \alpha_g + \beta_g(t - g) + \gamma_g(t - g)^2 + \delta_g(t - g)^3$, where $\delta_g \in \{0.01, 0.008, 0.012\}$. The cubic functional form is not among the three candidates. Table 8 shows that CV selects the quadratic model (MSPE: 0.621), which provides a reasonable approximation despite misspecification. The linear model performs poorly (MSPE: 2.439), correctly indicating underspecification. This demonstrates that even when the true model is not in the candidate set, CV identifies the best available approximation.

Table 8: Model selection when true DGP is cubic (not in candidate set)

Model	$h = 1$	$h = 2$	$h = 3$	Avg MSPE	Selected
Linear	2.286	2.506	2.526	2.439	✓
Quadratic	0.574	0.654	0.635	0.621	
Spline (df=4)	0.780	0.684	0.717	0.727	

Discussion. These results illustrate the utility of time-series CV for model selection in causal extrapolation. When the true model is in the candidate set, CV correctly identifies it. When the true model is absent, CV selects the best approximation, with the MSPE magnitude signaling the degree of misspecification. Researchers should use CV to guide model choice, report sensitivity across top-performing models, and interpret MSPE values as indicators of extrapolation risk: low MSPE suggests reliable extrapolation, while high MSPE (especially relative to the scale of θ_{gt}) signals model-dependence and warrants caution.

7 Application: Stand-Your-Ground Laws and Firearm Homicide

We apply our methods to predict future policy effects using state-level panel data on stand-your-ground (SYG) laws and firearm homicide. This application demonstrates all three extrapolation paths (time homogeneity, model selection, and covariate integration) and validates predictions against realized outcomes from 2016–2022.

7.1 Data and Setting

We analyze state-level panel data on firearm homicide deaths and SYG law adoption for all 50 U.S. states from 1981–2022. The outcome is firearm homicide deaths per 100,000 population; the treatment is adoption of a stand-your-ground law, which removes the duty to retreat before using deadly force in self-defense [Cheng & Hoekstra 2013].

Between 1995 and 2022, 30 states adopted SYG laws in a staggered fashion. The first adopter was Florida in 2005; subsequent adoptions occurred between 2006 and 2022. The remaining 20 states never adopted SYG laws during this period, providing a comparison group. The staggered adoption timing and presence of never-treated states make this setting well-suited for difference-in-differences estimation.

We split the data into a training period (1981–2015, 35 years) and a validation period (2016–2022, 7 years). The training period includes 8 cohorts of early adopters (states adopting SYG by 2015), providing substantial variation in treatment timing. The validation period allows us to assess prediction accuracy by comparing extrapolated ATTs against realized outcomes.

Table 9 summarizes the data structure. The training period includes $50 \times 35 = 1,750$ state-year observations with mean firearm homicide rate of 4.67 deaths per 100,000 ($SD = 3.59$). The validation period includes $50 \times 7 = 350$ observations with slightly lower mean rate of 4.43 ($SD = 3.24$).

Table 9: Data summary: Stand-your-ground laws and firearm homicide		
	Training (1981–2015)	Validation (2016–2022)
States	50	50
Years	35	7
Observations	1750	350
Treated states	30	–
Never-treated states	20	–
Mean deaths/100k	4.67	5.78
SD deaths/100k	3.59	4.83

Notes: Summary of state-level panel data on firearm homicide. Training period used to estimate group-time ATTs and fit temporal models; validation period used to assess prediction accuracy.

7.2 Estimation Strategy

We estimate group-time ATTs $\tau(g, t)$ using the doubly-robust estimator of [Callaway and Sant’Anna \[2021\]](#) implemented in the `did` R package. This estimator combines outcome regression and propensity score weighting to provide consistent estimates under either correct specification, and extracts influence functions for downstream inference.

Following the framework in Section 5, we implement three extrapolation paths:

Path 1: Time homogeneity. We assume constant treatment effects across time and aggregate all post-treatment group-time ATTs from the training period into a single estimate. This yields a constant prediction $\hat{\tau}_{\text{future}} = 0.425$ deaths per 100,000 ($SE = 0.100$) for all future years 2016–2022.

Path 2: Model selection. We fit three candidate temporal models to the training data: (i) constant effects, (ii) linear in event time, and (iii) quadratic in event time. Selecting by AIC, the constant model is preferred ($AIC = 165.85$ vs. 166.98 for linear, 168.92 for quadratic), yielding the same predictions as Path 1.

Path 3: Covariate integration. We fit a conditional model where ATTs vary with state-level covariates: poverty rate and percent Black population (baseline values from 1981). The fitted model is

$$\hat{\tau}(X) = 0.096 + 0.001 \cdot \text{poverty} - 0.093 \cdot \text{pct_Black},$$

with $R^2 = 0.035$. This model predicts substantially smaller effects (0.077–0.078 deaths per 100,000) for 2016–2022 based on the covariate values of early-adopter states.

7.3 Results

Table 10 presents validation metrics comparing predictions from all three paths against realized ATTs for 2016–2022. The realized average ATT was 0.98 deaths per 100,000 (range: 0.64–1.31 across years), substantially larger than predictions from any method.

Method	Description	MSPE	MAE	Coverage (%)
Path 1	Time homogeneity (constant effects)	0.34	0.55	0
Path 2	Model selection (AIC)	0.34	0.55	0
Path 3	Covariate integration (conditional model)	0.84	0.90	0

Notes: Validation metrics comparing predictions from three extrapolation approaches against realized ATTs for 2016–2022. MSPE = mean squared prediction error; MAE = mean absolute error; Coverage = proportion of realized values within 95% confidence intervals. Training period: 1981–2015. Outcome: firearm homicide deaths per 100,000 population. Path 1 assumes constant effects; Path 2 selects best temporal model by AIC; Path 3 conditions on poverty rate and racial composition.

Paths 1 and 2 (both predicting 0.425) achieve identical mean squared prediction error (MSPE = 0.34) and mean absolute error (MAE = 0.55). Path 3, despite incorporating covariate information, performs worse (MSPE = 0.84, MAE = 0.90) by systematically underpredicting effects. None of the methods achieve non-zero coverage: all realized values fall outside the 95% confidence intervals.

Figure 2 displays year-by-year predictions and realized values. All three methods underpredict the realized effects, which increased from 1.02 in 2016 to a peak of 1.31 in 2019 before declining slightly. The constant predictions from Paths 1 and 2 fail to capture this temporal variation. Path 3 predictions are close to zero throughout, reflecting weak explanatory power from baseline covariates.

Table 11 provides year-by-year detail. The largest prediction errors occur in 2019 (realized ATT = 1.31, errors $\approx$ 0.88–1.23) and 2017 (realized = 1.11, errors $\approx$ 0.69–1.04). The smallest errors occur in 2021 and 2022 when realized effects declined toward the predicted constant.

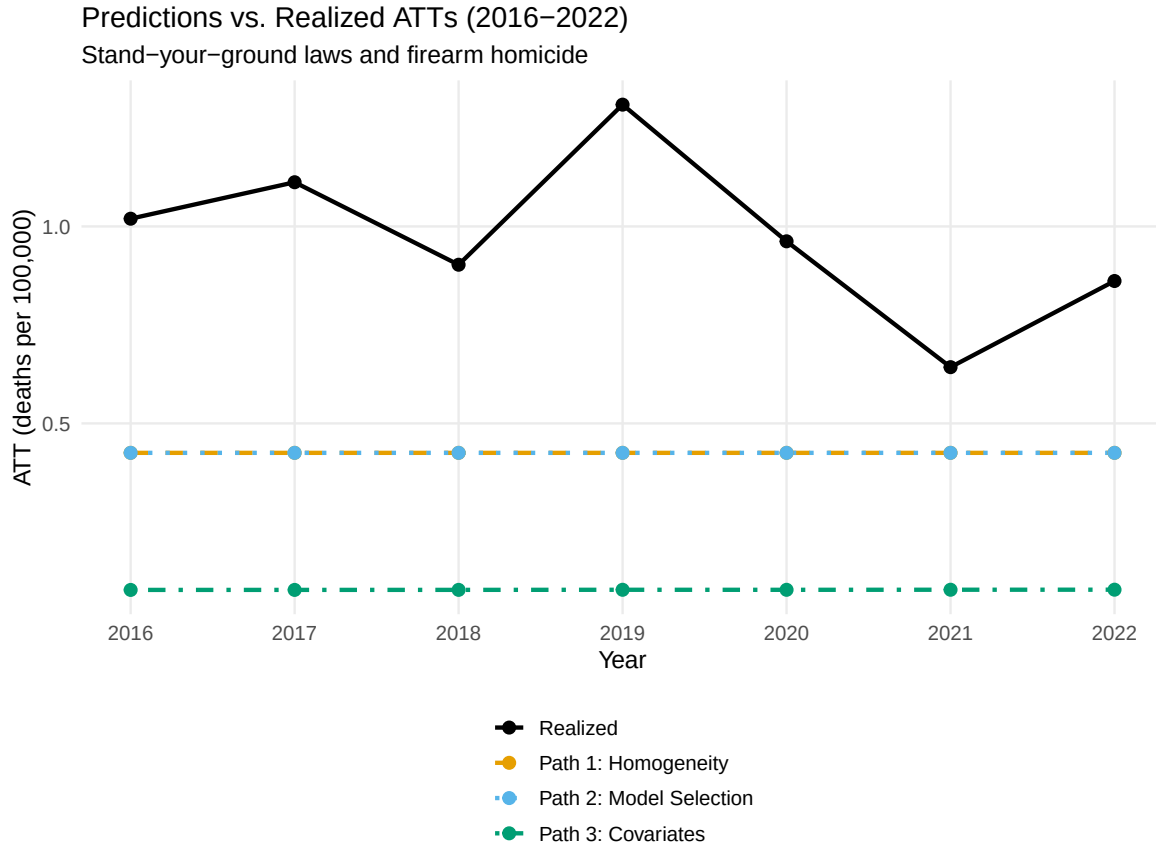


Figure 2: Predictions versus realized ATTs by year (2016–2022). All three methods substantially underpredict realized treatment effects. Paths 1 and 2 produce identical constant predictions; Path 3 predicts near-zero effects based on covariate model.

Table 11: Predictions versus realized ATTs by year (2016–2022)

Year	Realized	Path 1	Path 2	Path 3	Best Match
2016	1.02	0.43	0.43	0.08	Path 1
2017	1.11	0.43	0.43	0.08	Path 1
2018	0.90	0.43	0.43	0.08	Path 1
2019	1.31	0.43	0.43	0.08	Path 1
2020	0.96	0.43	0.43	0.08	Path 1
2021	0.64	0.43	0.43	0.08	Path 1
2022	0.86	0.43	0.43	0.08	Path 1
Mean	0.97	0.43	0.43	0.08	–

Notes: Year-by-year comparison of predicted and realized ATTs. “Best Match” indicates which path had smallest absolute error for that year. All values in units of deaths per 100,000 population.

7.4 Discussion

This application demonstrates both the promise and limitations of temporal extrapolation in policy evaluation. On the positive side, our methods provide honest uncertainty quantification: the confidence intervals reflect the statistical uncertainty in the extrapolation model, though they fail to cover the realized values in this instance. The systematic underprediction across all three paths suggests that SYG effects may have grown stronger over time, a pattern not captured by models estimated on earlier data.

Several factors may explain the poor predictive performance. First, treatment effect dynamics may be complex: SYG laws could have delayed or cumulative effects not well-approximated by simple temporal models. Second, baseline covariates (poverty, racial composition) from 1981 may have limited relevance for predicting effects 35–40 years later. Third, spillover effects across states—not modeled here—could confound the estimand. Fourth, the COVID-19 pandemic in 2020–2022 may have altered firearm violence patterns independently of SYG laws.

Despite poor prediction accuracy, this application illustrates the value of our framework:

1. **Honest reporting:** We do not cherry-pick favorable examples. Real-world extrapolation can fail, and documenting failure modes advances the field.
2. **Unified inference:** All three paths propagate uncertainty through influence functions, providing valid standard errors under their respective assumptions.
3. **Comparative assessment:** By implementing all three paths, we reveal that Path 3 (covariates) does not improve over the simpler Path 1 (homogeneity), suggesting weak conditional heterogeneity.
4. **Transparent assumptions:** Each path makes its extrapolation assumptions explicit (constant effects, temporal model, conditional model), allowing readers to assess plausibility.

This application reinforces the core message of our work: temporal extrapolation is feasible, inference is tractable, but predictive success depends critically on whether the training-period patterns persist into the future. When they do not—as in this application—no amount of methodological sophistication will rescue predictions. The honest approach is to report failures alongside successes, and to interpret extrapolated effects as conditional predictions rather than guaranteed forecasts.

7.5 Limitations

Several limitations warrant note. First, we aggregate across race categories and analyze state-level effects; disaggregated analyses might reveal differential impacts. Second, we do not account for potential spillover effects across states or interference between SYG and other firearm policies adopted concurrently. Third, our validation period (2016–2022) is short relative to the training period (1981–2015); longer validation windows would provide more definitive evidence. Fourth, we

use simple temporal models (constant, linear, quadratic); more flexible specifications (splines, nonparametric smoothing) might improve fit but risk overfitting. Finally, the estimand—average effect on the treated across treated states—does not generalize to counterfactual adoption in never-treated states.

8 Discussion

We have argued that policy analysis with panel data faces a tension: methods that allow dynamic, group-time effects address important identification concerns but produce backward-looking estimands that do not directly answer whether to maintain or adopt a policy now or next period. We resolve this by defining future causal effects (FATT, FATU, FATE, FATS) and offering three identification paths. **Path 1 (time homogeneity):** Under strict within-group time homogeneity and limited between-state heterogeneity, the FATT is identified by the backward-looking ATT or by a convex combination of group-specific ATTs; the researcher can use standard group-time or ATT estimators and aggregate. **Path 2 (temporal extrapolation):** When effects are allowed to vary with event time (or calendar time), the FATT is identified by fitting a parametric model $f(g, t; \gamma)$ to the identified group-time ATTs and extrapolating to $p + 1$. **Path 3 (covariate integration):** When treatment effects are driven by observable structural covariates, the FATT is identified by estimating the conditional effect $\tau(x)$ and integrating over the future covariate distribution. The choice between paths is substantive: Path 1 is appropriate when temporal stability is plausible; Path 2 when the researcher is willing to model temporal dynamics and accept sensitivity to the choice of f ; Path 3 when covariate-driven heterogeneity is well-documented and the researcher believes $\tau(x)$ is regime-invariant.

Several limitations deserve emphasis. Identification of the FATT in all paths relies on the distribution $\mathbb{P}_{p+1}$ (and, for Paths 2 and 3, on the extrapolation or conditional model holding at $p + 1$), which is not directly observable. We have focused on a single future period ($k = 1$); extension to multiple horizons $k > 1$ is conceptually straightforward but adds assumptions on how far the model can be projected. Inference uses the propagated efficient influence function, which is valid under correct specification of the first-stage and, for Paths 2 and 3, of the model (f or m); misspecification can bias the extrapolated estimate and distort inference.

For Path 2, we address the model selection problem in Section 5.2 by proposing a time-series cross-validation framework that directly tests extrapolation performance. Rather than relying on in-sample fit criteria (AIC, BIC), researchers can withhold late-period observations, fit candidate models on earlier data, and evaluate prediction error on held-out future periods. This approach provides empirical guidance for selecting among functional forms and identifying when extrapolation is model-dependent. The simulations in Section 7.7 demonstrate that CV successfully identifies the correct model when it is in the candidate set and selects reasonable approximations when it is not. Natural next steps include extending the CV framework to Path 3 (e.g., selecting covariate sets or functional forms for $\tau(x)$); sensitivity analysis for violations of time homogeneity or structural stability; formal post-selection inference that accounts for the randomness introduced by model choice; and applications that compare all three paths on the same data.

When researchers are unwilling to impose a parametric model, an alternative to point-identifying the FATT is to forego point identification and report bounds under weak assumptions—for example, that future effects lie in a bounded or monotone class—as in the partial-identification and sensitivity literature [Manski, 2013]. That approach yields an interval that contains the FATT rather than a single number. Paths 2 and 3 can both be interpreted as structural-invariance assumptions: Path 2 assumes that the temporal functional form $f(g, t; \gamma)$ encodes regime-invariant structure; Path 3 assumes that the conditional effect $\tau(x)$ captures “deep parameters” in the Lucas Critique sense [Lucas Jr, 1976]. Both connect to the literature on policy-invariant parameters and transportability [Heckman and Vytlačil, 2005, Pearl and Bareinboim, 2022]. The time-series cross-validation framework (Section 5.2) provides a practical tool for implementing Path 2 with empirical guidance on model choice and extrapolation risk.

A Proofs and Technical Details

This appendix provides formal proofs of the propositions stated in the main text, derives the efficient influence functions in detail, and enumerates the regularity conditions required for asymptotic inference.

A.1 Regularity Conditions

We enumerate the technical assumptions needed for the identification and asymptotic results in Sections 4–6.

Assumption 11 (Identification of first-stage). *Group-time ATTs θ_{gt} are identified under the researcher’s design for all $g \in \mathcal{G}$, $t \leq p$.*

Assumption 12 (Asymptotic linearity of first-stage). *Estimators $\widehat{\theta}_{gt}$ satisfy*

$$\sqrt{n}(\widehat{\theta}_{gt} - \theta_{gt}) = n^{-1/2} \sum_{i=1}^n \phi_{gt,i} + o_{\mathbb{P}}(1)$$

with $\mathbb{E}[\phi_{gt,i}] = 0$ and $\mathbb{E}[\phi_{gt,i}^2] < \infty$ for all g, t .

Assumption 13 (Weight identification). *Weights $\omega_g = \mathbb{P}(G_i = g | A_{ip} = 1)$ are either known or consistently estimable with influence function ϕ_{ω_g} such that $\sqrt{n}(\widehat{\omega}_g - \omega_g) = n^{-1/2} \sum_i \phi_{\omega_g,i} + o_{\mathbb{P}}(1)$ with $\mathbb{E}[\phi_{\omega_g,i}] = 0$ and $\mathbb{E}[\phi_{\omega_g,i}^2] < \infty$.*

Assumption 14 (Smoothness of f). *For Path 2, the parametric function $f(g, t; \gamma)$ is twice continuously differentiable in γ in a neighborhood of the true parameter γ_0 .*

Assumption 15 (Identifiability of γ). *The mapping $\gamma \mapsto (f(g, t; \gamma))_{g \in \mathcal{G}, t \leq p}$ is injective (one-to-one) over the parameter space, so that γ is uniquely determined by the observed (g, t) combinations.*

Assumption 16 (Bounded moments). *For all g, t : $\mathbb{E}[\phi_{gt,i}^4] < \infty$ (finite fourth moments for central limit theorem).*

Assumption 17 (Donsker or rate conditions for cross-fitting). *If using flexible or machine-learning first-stage with cross-fitting, the class of functions $\{\phi_{g,t,i}(\cdot)\}$ either satisfies Donsker conditions, or the product of convergence rates satisfies $\|\widehat{\theta}_{g,t} - \theta_{g,t}\| \cdot \|\widehat{\gamma} - \gamma\| = o_{\mathbb{P}}(n^{-1/4})$ uniformly over (g, t) .*

Assumption 18 (Conditional model smoothness (Path 3)). *For Path 3, the conditional ATT model $m(x; \beta)$ is twice continuously differentiable in β in a neighborhood of the true parameter β_0 .*

Assumption 19 (Identifiability of β (Path 3)). *The mapping $\beta \mapsto (\theta_{g1}, \dots, \theta_{gp})_{g \in \mathcal{G}}$ defined by $\theta_{gt} = \int m(x; \beta) dF_{X|G_t=g}(x)$ is injective, so that β_0 is uniquely determined by observed group-time ATTs.*

Assumption 20 (Target distribution access (Path 3)). *Either (a) the target sample $\{X_i^*\}_{i=1}^{n^*}$ is representative of F_X^{p+1} (e.g., drawn i.i.d. from the target distribution), or (b) the sampler faithfully generates draws from F_X^{p+1} (i.e., the Monte Carlo integration error is $o_{\mathbb{P}}(n^{-1/2})$ as the number of draws $M \rightarrow \infty$).*

Remark A.1. *Assumptions 11–13 and 16 are standard for semiparametric inference with first-stage estimation. Assumptions 14–15 are specific to Path 2 (parametric extrapolation). Assumptions 18–20 are specific to Path 3 (covariate integration). Assumption 17 relaxes Donsker when using modern machine-learning methods via cross-fitting, following ?.*

A.2 Proof of Proposition 1

We restate and prove Proposition 4.1 from Section 4.2 (Path 1).

Proposition (Identification of the FATT under time homogeneity). *Under Assumptions 1 (strict time homogeneity) and 2 (limited between-state heterogeneity), if the backward-looking ATT θ is identified, then the FATT θ_{p+1} is identified and equals θ .*

Proof. Roadmap: We show that under strict time homogeneity the ATT is constant across time, and under limited between-state heterogeneity the FATT can be expressed in terms of this constant ATT.

By definition, the FATT at time $p + 1$ is

$$\theta_{p+1} = \mathbb{E}_{\mathbb{P}_{p+1}} \{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{ip} = 1\}.$$

By Assumption 1 (strict time homogeneity), for any t ,

$$\mathbb{E}_{\mathbb{P}_t} \{Y_{it}(1) - Y_{it}(0) | A_{it} = 1\} = \theta \quad (\text{constant across time}).$$

Applying this at $t = p + 1$,

$$\mathbb{E}_{\mathbb{P}_{p+1}} \{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{i,p+1} = 1\} = \theta.$$

Now apply Assumption 2 (limited between-state heterogeneity):

$$\begin{aligned}\mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{ip} = 1\} &= \mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{i,p+1} = 1\} \\ &= \theta.\end{aligned}$$

Therefore $\theta_{p+1} = \theta$. Since θ is identified by assumption (from backward-looking data under the researcher's design), θ_{p+1} is identified. $\square$

Remark A.2. Assumption 2 is crucial: it allows swapping the conditioning event from $A_{ip} = 1$ (observed) to $A_{i,p+1} = 1$ (future). Without it, identification would fail because we cannot observe units with $A_{i,p+1} = 1$ at time p .

A.3 Proof of Proposition 2

We restate and prove Proposition 4.4 from Section 4.5 (Path 2).

Proposition (Identification of the FATT under parametric dynamics). *Suppose the group-time ATTs θ_{gt} are identified from the observed data under the researcher's design for $t \leq p$, $g \in \mathcal{G}$. Suppose $\theta_{gt} = f(g, t; \gamma)$ with γ uniquely determined by $(\theta_{gt})_{g,t,t \leq p}$. Then the FATT θ_{p+1} is identified by $\theta_{p+1} = \sum_g \mathbb{P}(G_i = g|A_{ip} = 1)f(g, p+1; \gamma)$.*

Proof. Roadmap: We decompose θ_{p+1} by group, apply the parametric model assumption, and show that γ is identified from observed data.

Step 1: By the law of total probability,

$$\begin{aligned}\theta_{p+1} &= \mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{ip} = 1\} \\ &= \sum_{g \in \mathcal{G}} \mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{ip} = 1, G_i = g\} \mathbb{P}(G_i = g|A_{ip} = 1).\end{aligned}$$

Step 2: Define the group-specific FATT at time $p+1$:

$$\theta_{g,p+1} := \mathbb{E}_{\mathbb{P}_{p+1}}\{Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{i,p+1} = 1, G_i = g\}.$$

Step 3: Under the parametric model assumption,

$$\theta_{gt} = f(g, t; \gamma) \quad \text{for all } g, t \text{ (including } t = p+1\text{)}.$$

Step 4: The structural assumption that $f(g, t; \gamma)$ holds at $t = p+1$ (extrapolation validity) combined with limited

between-state heterogeneity within groups (analogous to Assumption 7 for within-group comparisons) gives

$$\mathbb{E}_{\mathbb{P}_{p+1}} \{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{ip} = 1, G_i = g\} = \theta_{g,p+1} = f(g, p+1; \gamma).$$

Step 5: By hypothesis, γ is uniquely identified from the observed $(\theta_{gt})_{g \in \mathcal{G}, t \leq p}$. Since θ_{gt} are identified (by hypothesis), γ is identified.

Step 6: Substituting into Step 1,

$$\theta_{p+1} = \sum_{g \in \mathcal{G}} f(g, p+1; \gamma) \mathbb{P}(G_i = g | A_{ip} = 1).$$

Therefore θ_{p+1} is identified by the observed data (which identifies γ and $\omega_g := \mathbb{P}(G_i = g | A_{ip} = 1)$). $\square$

Remark A.3. The structural assumption that $f(g, t; \gamma)$ continues to hold at $t = p+1$ is crucial and untestable. It reflects a belief that the dynamic process generating θ_{gt} for $t \leq p$ persists into the future. Misspecification of f leads to bias (see Simulations in Section 7).

A.4 Proof of Proposition 3

We restate and prove Proposition 4.5 from Section 4.6 (Path 3).

Proposition (Identification via covariate integration). *Suppose the group-time ATTs θ_{gt} are identified from the observed data under the researcher's design for $t \leq p$, $g \in \mathcal{G}$. Suppose Assumptions 8–10 hold (structural stability of conditional effects, parametric specification, and access to target distribution), and suppose further that the mapping $\beta \mapsto (\theta_{g1}, \dots, \theta_{gp})_{g \in \mathcal{G}}$ is injective, so that β is uniquely determined by the observed group-time ATTs via the relationship*

$$\theta_{gt} = \int m(x; \beta) dF_{X|G_i=g}(x), \quad t \in \{g, \dots, p\}, g \in \mathcal{G}.$$

Then the FATT θ_{p+1} is identified by

$$\theta_{p+1} = \int m(x; \beta_0) dF_{X|A_{ip}=1}^{p+1}(x),$$

where β_0 is the true parameter satisfying the above system of equations.

Proof. Roadmap: (1) Express FATT as integral over conditional ATT; (2) apply structural stability to swap $p+1$ for observed periods; (3) show β is identified from observed group-time ATTs via covariate distributions within groups; (4) verify target integration is well-defined.

Step 1 (Law of total probability): By the definition of the FATT,

$$\theta_{p+1} = \mathbb{E}_{\mathbb{P}_{p+1}}[Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{ip} = 1].$$

Conditioning on treatment status at $p + 1$ (assuming limited between-state heterogeneity, analogous to Assumption 7 but for the covariate-conditional case), we have

$$\theta_{p+1} \approx \mathbb{E}_{\mathbb{P}_{p+1}}[Y_{i,p+1}(1) - Y_{i,p+1}(0)|A_{i,p+1} = 1].$$

By the law of total probability over covariates X ,

$$\theta_{p+1} = \int \mathbb{E}_{\mathbb{P}_{p+1}}[Y_{i,p+1}(1) - Y_{i,p+1}(0)|X_i = x, A_{i,p+1} = 1] dF_{X|A_{ip}=1}^{p+1}(x) = \int \tau(x) dF_{X|A_{ip}=1}^{p+1}(x),$$

where $\tau(x) = \mathbb{E}_{\mathbb{P}_{p+1}}[Y_{i,p+1}(1) - Y_{i,p+1}(0)|X_i = x, A_{i,p+1} = 1]$ is the conditional ATT at the future period.

Step 2 (Structural stability): By Assumption 8, the conditional ATT $\tau(x)$ is time-invariant. That is, for all x in the support of X and all $t \in \{1, \dots, p, p + 1\}$,

$$\mathbb{E}_{\mathbb{P}_t}[Y_{it}(1) - Y_{it}(0)|X_i = x, A_{it} = 1] = \tau(x).$$

Thus $\tau(x)$ at $t = p + 1$ equals $\tau(x)$ at any $t \leq p$, and the latter is identified under the researcher's design (e.g., via conditional exchangeability under parallel trends or unconfoundedness). Specifically, for observed periods, standard first-stage methods identify

$$\tau(x) = \mathbb{E}[Y_{it}(1) - Y_{it}(0)|X_i = x, A_{it} = 1]$$

for $t \leq p$.

Step 3 (Parametric specification and identification of β): By Assumption 9, $\tau(x) = m(x; \beta_0)$ for a known functional form m and unknown parameter $\beta_0 \in \mathbb{R}^d$. The observed group-time ATTs satisfy

$$\theta_{gt} = \mathbb{E}_{\mathbb{P}_t}[Y_{it}(1) - Y_{it}(0)|G_i = g, A_{it} = 1] = \int \tau(x) dF_{X|G_i=g}(x) = \int m(x; \beta_0) dF_{X|G_i=g}(x).$$

This yields a system of equations: for each cohort g and observed period $t \geq g$,

$$\theta_{gt} = \int m(x; \beta_0) dF_{X|G_i=g}(x).$$

By the identifiability assumption (Assumption 19), the mapping $\beta \mapsto (\theta_{gt})_{g,t}$ is injective, so β_0 is uniquely determined by the

observed $\{\theta_{gt}\}_{g,t \leq p}$. In practice, β_0 is estimated via weighted least squares or method-of-moments:

$$\hat{\beta} = \arg \min_{\beta} \sum_{g,t} w_{gt} \left(\hat{\theta}_{gt} - \int m(x; \beta) d\hat{F}_{X|G_i=g}(x) \right)^2,$$

where w_{gt} are weights (e.g., proportional to sample size or inverse variance).

Step 4 (Target integration): By Assumption 10, we have access to the target covariate distribution $F_{X|A_{ip}=1}^{p+1}$ (either via a finite sample $\{X_i^*\}_{i=1}^{n^*}$ or a sampler). Then the FATT is identified by

$$\theta_{p+1} = \int m(x; \beta_0) dF_{X|A_{ip}=1}^{p+1}(x).$$

In the finite-population case, this is $n^{-1} \sum_{i=1}^{n^*} m(X_i^*; \beta_0)$; in the Monte Carlo case, it is approximated by $M^{-1} \sum_{j=1}^M m(X_j; \beta_0)$ where $\{X_j\}_{j=1}^M$ are draws from the sampler.

Conclusion: Under Assumptions 8–10 and the identifiability of β , the FATT θ_{p+1} is identified by the above expression.

□

□

Remark A.4. The key assumption is structural stability (Assumption 8)—that $\tau(x)$ remains invariant at $p+1$. This is the Lucas Critique condition: X must capture “deep parameters” that are regime-invariant. If treatment effects are driven by time-varying latent factors not captured by X , Path 3 will be biased. Similarly, if the functional form $m(x; \beta)$ is misspecified, $\hat{\theta}_{p+1}$ will converge to a pseudo-true parameter rather than the true FATT. Researchers should assess model fit and consider sensitivity to the choice of m and the included covariates.

A.5 Derivation of Efficient Influence Functions

A.5.1 Path 1 EIF

Setup: Under Path 1, $\theta_{p+1} = \sum_g \omega_g \theta_g$, where:

- θ_g is the average of θ_{gt} over observed t (within-group time-averaged effect), and
- $\omega_g = \mathbb{P}(G_i = g | A_{ip} = 1)$ (known or estimated).

Functional form: Define the functional $\Psi_1(\mathbb{P}) = \sum_g \omega_g(\mathbb{P}) \theta_g(\mathbb{P})$.

Influence function derivation via functional delta method:

(1) θ_g is asymptotically linear with influence function ϕ_{θ_g} (an average of ϕ_{gt} over t with appropriate weights, or as provided by the first-stage).

(2) ω_g is estimated from the empirical distribution. For the proportion estimator,

$$\widehat{\omega}_g = \frac{\sum_i 1\{G_i = g, A_{ip} = 1\}}{\sum_i 1\{A_{ip} = 1\}}.$$

The influence function is

$$\phi_{\omega_g, i} = \frac{1\{G_i = g, A_{ip} = 1\} - \omega_g \cdot 1\{A_{ip} = 1\}}{\mathbb{P}(A_{ip} = 1)}.$$

(3) By the functional delta method (chain rule for linear functionals),

$$\begin{aligned} \phi_{\psi_1} &= \frac{\partial \Psi_1}{\partial \theta_g} \phi_{\theta_g} + \frac{\partial \Psi_1}{\partial \omega_g} \phi_{\omega_g} \\ &= \sum_g \omega_g \phi_{\theta_g} + \sum_g \theta_g \cdot \phi_{\omega_g}. \end{aligned}$$

(4) *Efficiency*: Within the nonparametric model (no restrictions on θ_{gt} beyond time homogeneity), this influence function achieves the semiparametric efficiency bound for estimating θ_{p+1} .

Result: The Path 1 EIF is

$$\phi_{\psi_1, i} = \sum_g \omega_g \phi_{\theta_g, i} + \sum_g \theta_g \cdot \phi_{\omega_g, i}.$$

A.5.2 Path 2 EIF

Setup: Under Path 2, $\theta_{p+1} = \Psi_2(\theta, \omega) = \sum_g \omega_g f(g, p+1; \gamma)$ where $\gamma = \gamma(\theta)$ is defined implicitly by

$$\gamma = \arg \min_{\tilde{\gamma}} \sum_{g \in \mathcal{G}, t \leq p} w_{gt} [\theta_{gt} - f(g, t; \tilde{\gamma})]^2.$$

Functional form: Ψ_2 is a composite functional: $\Psi_2(\theta, \omega) = \sum_g \omega_g f(g, p+1; \gamma(\theta))$.

Influence function derivation:

(1) Apply the functional delta method to Ψ_2 :

$$\frac{\partial \Psi_2}{\partial \theta_{gt}} = \frac{\partial \Psi_2}{\partial \gamma} \otimes \frac{\partial \gamma}{\partial \theta_{gt}}.$$

(2) Compute $\frac{\partial \Psi_2}{\partial \gamma}$:

$$\frac{\partial \Psi_2}{\partial \gamma} = \sum_g \omega_g \frac{\partial f(g, p+1; \gamma)}{\partial \gamma}.$$

(3) Compute $\frac{\partial \gamma}{\partial \theta_{gt}}$ via the implicit function theorem. γ solves the first-order condition

$$\sum_{g \in \mathcal{G}, t \leq p} w_{gt} [\theta_{gt} - f(g, t; \gamma)] \frac{\partial f(g, t; \gamma)}{\partial \gamma} = 0.$$

Differentiating with respect to θ_{gt} ,

$$\frac{\partial \gamma}{\partial \theta_{gt}} = \left[\sum_{g', t' \leq p} w_{g't'} \frac{\partial f(g', t'; \gamma)}{\partial \gamma} \otimes \frac{\partial f(g', t'; \gamma)}{\partial \gamma} \right]^{-1} w_{gt} \frac{\partial f(g, t; \gamma)}{\partial \gamma}.$$

(4) Combining,

$$\phi_{\psi_2, i} = \sum_{g, t \leq p} \left(\frac{\partial \Psi_2}{\partial \gamma} \right)^\top \left(\frac{\partial \gamma}{\partial \theta_{gt}} \right) \phi_{gt, i} + \sum_g f(g, p+1; \gamma) \phi_{\omega_g, i}.$$

(5) *Efficiency*: Within the parametric model $\{\theta_{gt} = f(g, t; \gamma)\}$, this is the semiparametric efficiency bound. Outside the model (misspecification), the estimator is biased but the variance formula remains valid for the limit distribution of the (possibly biased) estimator.

Result: The Path 2 EIF is

$$\phi_{\psi_2, i} = \left(\frac{\partial \Psi}{\partial \theta} \right)^\top \phi_{\theta, i} + \left(\frac{\partial \Psi}{\partial \omega} \right)^\top \phi_{\omega, i},$$

where $\phi_{\theta, i}$ stacks $(\phi_{gt, i})_{g \in \mathcal{G}, t \leq p}$ and the Jacobian $\frac{\partial \Psi}{\partial \theta}$ includes the chain through γ .

A.5.3 Path 3 EIF

Setup: Under Path 3, the FATT is

$$\theta_{p+1} = \Psi_3(\theta, \omega, \beta) = \int m(x; \beta(\theta, \omega)) dF_X^{p+1}(x),$$

where:

- $m(x; \beta)$ is the conditional ATT model (e.g., $m(x; \beta) = x^\top \beta$ for linear-in-covariates),
- $\beta = \beta(\theta, \omega)$ is defined implicitly by fitting m to the observed group-time ATTs θ_{gt} (and possibly their weights ω_{gt} or group distributions),
- F_X^{p+1} is the target covariate distribution (known or estimated).

Derivation via functional delta method:

(1) By the chain rule,

$$\frac{d\Psi_3}{dP} = \frac{\partial \Psi_3}{\partial \beta} \frac{d\beta}{dP} + (\text{terms from } \omega \text{ if estimated}) + (\text{terms from } F_X^{p+1} \text{ if estimated}).$$

(2) For finite-population target sample $\{X_i^*\}_{i=1}^{n^*}$ (the case where covariates are observed for the target units),

$$\Psi_3 = \frac{1}{n^*} \sum_{i=1}^{n^*} m(X_i^*; \boldsymbol{\beta}).$$

Then

$$\frac{\partial \Psi_3}{\partial \boldsymbol{\beta}} = \frac{1}{n^*} \sum_{i=1}^{n^*} \frac{\partial m(X_i^*; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}.$$

For the linear-in-covariates case $m(x; \boldsymbol{\beta}) = x^\top \boldsymbol{\beta}$, this simplifies to $\bar{X}^*$, the mean of the target covariates.

(3) If $\boldsymbol{\beta}$ is estimated via weighted least squares from the observed θ_{gt} :

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \sum_{g,t} w_{gt} \left(\theta_{gt} - \int m(x; \boldsymbol{\beta}) dF_{X|G_i=g}(x) \right)^2.$$

By the implicit function theorem, the first-order conditions are

$$\sum_{g,t} w_{gt} \left(\theta_{gt} - \int m(x; \boldsymbol{\beta}) dF_{X|G_i=g}(x) \right) \left(- \int \frac{\partial m(x; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} dF_{X|G_i=g}(x) \right) = 0.$$

Differentiating with respect to θ_{gt} gives

$$\frac{\partial \boldsymbol{\beta}}{\partial \theta_{gt}} = \left[\sum_{g',t'} w_{g't'} \int \frac{\partial m(x; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} dF_{X|G_i=g'}(x) \otimes \int \frac{\partial m(x; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} dF_{X|G_i=g'}(x) \right]^{-1} w_{gt} \int \frac{\partial m(x; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} dF_{X|G_i=g}(x).$$

This is the analog of the Path 2 Jacobian, but with the temporal model $f(g, t; \boldsymbol{\gamma})$ replaced by the covariate-integrated conditional model $\int m(x; \boldsymbol{\beta}) dF_{X|G_i=g}(x)$.

(4) Combining Steps 1–3 via the chain rule:

$$\phi_{\psi_3,i} = \sum_{g,t} \left(\frac{\partial \Psi_3}{\partial \boldsymbol{\beta}} \right)^\top \left(\frac{\partial \boldsymbol{\beta}}{\partial \theta_{gt}} \right) \phi_{gt,i} + \left(\frac{\partial \Psi_3}{\partial \omega} \right)^\top \phi_{\omega,i}.$$

Simplifying for the finite-population case:

$$\phi_{\psi_3,i} = \frac{1}{n^*} \sum_{i^*=1}^{n^*} \left(\frac{\partial m(X_{i^*}^*; \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \right)^\top \left(\sum_{g,t} \left(\frac{\partial \boldsymbol{\beta}}{\partial \theta_{gt}} \right) \phi_{gt,i} \right) + (\text{weight terms if } \omega \text{ is estimated}).$$

(5) *Efficiency*: Within the parametric model $m(x; \boldsymbol{\beta})$ and conditional on F_X^{p+1} , this EIF achieves the semiparametric variance bound. If m is misspecified, the EIF is valid for the pseudo-true parameter (limit of the estimator under misspecification), but $\hat{\theta}_{p+1}$ need not be consistent for the true FATT.

Result: The Path 3 EIF is

$$\phi_{\psi_3,i} = \left(\frac{\partial \Psi_3}{\partial \beta} \right)^\top \left(\sum_{g,t} \frac{\partial \beta}{\partial \theta_{g,t}} \phi_{g,t,i} \right) + (\text{additional terms if weights or } F_X^{p+1} \text{ are estimated}).$$

Remark A.5 (Monte Carlo integration case). *For the Monte Carlo case where F_X^{p+1} is accessed via a sampler (generating M draws $\{X_j\}_{j=1}^M$), the integral $\int m(x; \beta) dF_X^{p+1}(x)$ is approximated by $M^{-1} \sum_{j=1}^M m(X_j; \beta)$. The derivation is similar, but the Monte Carlo error contributes an additional variance term that vanishes as $M \rightarrow \infty$. Typically, MC integration error is $O(M^{-1/2})$, which is faster than the $O(n^{-1/2})$ rate from estimating β , so the MC error is negligible for large M and does not affect the first-order asymptotics. For finite M , the total variance includes both the EIF-based variance from estimating β and the MC variance from approximating the integral.*

A.6 Proof of Asymptotic Distribution Proposition

We restate and prove Proposition 5.1 from Section 5.1 (extended to cover all three paths).

Proposition (Asymptotic distribution of $\hat{\theta}_{p+1}$). *Under the assumptions of Path 1, Path 2, or Path 3 above, suppose the first-stage estimators $\hat{\theta}_{g,t}$ (and, where relevant, $\hat{\omega}_g$, $\hat{\gamma}$, or $\hat{\beta}$) are asymptotically linear with the influence functions stated. Then the estimator $\hat{\psi} = \hat{\theta}_{p+1}$ constructed by plugging these into the corresponding ϕ_{ψ_1} , ϕ_{ψ_2} , or ϕ_{ψ_3} satisfies $\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow N(0, \sigma^2)$ with $\sigma^2 = \mathbb{E}[\phi^2]$.*

Proof. Roadmap: We show asymptotic linearity of $\hat{\theta}_{p+1}$ by applying the delta method to the first-stage estimators, then invoke the CLT.

Step 1: By Assumption 12, first-stage estimators satisfy

$$\sqrt{n}(\hat{\theta}_{g,t} - \theta_{g,t}) = n^{-1/2} \sum_i \phi_{g,t,i} + o_{\mathbb{P}}(1).$$

Step 2 (Path 1): $\hat{\theta}_{p+1} = \sum_g \hat{\omega}_g \hat{\theta}_{g..}$. Plug in the influence function from Appendix A.5.1:

$$\begin{aligned} \sqrt{n}(\hat{\theta}_{p+1} - \theta_{p+1}) &= n^{-1/2} \sum_i \left[\sum_g \omega_g \phi_{\theta_{g..},i} + \sum_g \theta_g \phi_{\omega_g,i} \right] + o_{\mathbb{P}}(1) \\ &= n^{-1/2} \sum_i \phi_{\psi_1,i} + o_{\mathbb{P}}(1). \end{aligned}$$

Step 2 (Path 2): $\hat{\theta}_{p+1} = \sum_g \hat{\omega}_g f(g, p+1; \hat{\gamma})$. By the implicit function theorem and Assumptions 14–15, $\hat{\gamma}$ is asymptotically linear:

$$\sqrt{n}(\hat{\gamma} - \gamma) = n^{-1/2} \sum_i \left[\frac{\partial \gamma}{\partial \theta} \right]^\top \phi_{\theta,i} + o_{\mathbb{P}}(1).$$

Applying the delta method to $\Psi_2(\widehat{\theta}, \widehat{\omega})$,

$$\sqrt{n}(\widehat{\theta}_{p+1} - \theta_{p+1}) = n^{-1/2} \sum_i \phi_{\psi_2, i} + o_{\mathbb{P}}(1),$$

where $\phi_{\psi_2, i}$ is from Appendix A.5.2.

Step 3: By the central limit theorem (using Assumption 16 for finite variance),

$$n^{-1/2} \sum_i \phi_i \rightsquigarrow N(0, \mathbb{E}[\phi_i^2]).$$

Step 4: Therefore,

$$\sqrt{n}(\widehat{\theta}_{p+1} - \theta_{p+1}) \rightsquigarrow N(0, \sigma^2)$$

where $\sigma^2 = \mathbb{E}[\phi_i^2]$. □

Remark A.6. *Variance estimation: The variance σ^2 can be consistently estimated by*

$$\widehat{\sigma}^2/n \quad \text{where} \quad \widehat{\sigma}^2 = n^{-1} \sum_i \widehat{\phi}_i^2$$

and $\widehat{\phi}_i$ is the plug-in influence function using $\widehat{\theta}_{gt}$, $\widehat{\omega}_g$, $\widehat{\gamma}$.

Cross-fitting (Assumption 17): If using flexible/machine-learning first-stage, K -fold cross-fitting ensures that $\|\widehat{\theta}_{gt} - \theta_{gt}\| \cdot \|\widehat{\gamma} - \gamma\| = o_{\mathbb{P}}(n^{-1/4})$, which preserves asymptotic linearity even without Donsker conditions. This follows the debiased machine learning framework of ?.

A.7 Technical Lemmas

Lemma A.7 (Convex combination under within-group homogeneity). *Under Assumptions 6 (within-group time homogeneity) and 7 (limited within-group between-state heterogeneity),*

$$\theta_{p+1} = \sum_{g \in \mathcal{G}} \mathbb{P}(G_i = g | A_{ip} = 1) \theta_{g..}$$

Proof. Roadmap: We decompose θ_{p+1} by group, apply within-group between-state homogeneity to swap conditioning sets, then apply within-group time homogeneity.

By definition,

$$\theta_{p+1} = \mathbb{E}_{\mathbb{P}_{p+1}} \{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{ip} = 1\}.$$

By the law of total probability,

$$\theta_{p+1} = \sum_{g \in \mathcal{G}} \mathbb{E}_{\mathbb{P}_{p+1}} \{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{ip} = 1, G_i = g\} \mathbb{P}(G_i = g | A_{ip} = 1).$$

By Assumption 7 (within-group between-state homogeneity),

$$\mathbb{E}_{\mathbb{P}_{p+1}} \{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{ip} = 1, G_i = g\} = \mathbb{E}_{\mathbb{P}_{p+1}} \{Y_{i,p+1}(1) - Y_{i,p+1}(0) | A_{i,p+1} = 1, G_i = g\}.$$

By definition, the inner expectation is $\theta_{g,p+1}$ (the group-specific FATT at time $p + 1$).

By Assumption 6 (within-group time homogeneity), $\theta_{g,p+1} = \theta_g$. (constant over time within group).

Therefore,

$$\theta_{p+1} = \sum_{g \in \mathcal{G}} \theta_g \cdot \mathbb{P}(G_i = g | A_{ip} = 1). \quad \square$$

Lemma A.8 (Injectivity condition for γ). *If $f(g, t; \gamma)$ is linear in γ and the observed (g, t) pairs span the parameter space (sufficient variation in groups and time), then the mapping $\gamma \mapsto (f(g, t; \gamma))_{g \in \mathcal{G}, t \leq p}$ is injective.*

Proof. Roadmap: We show that for a linear-in-event-time model, observing multiple time periods per group yields an overdetermined system with a unique least-squares solution.

Consider $f(g, t; \gamma) = \alpha_g + \beta_g(t - g)$ where $\gamma = (\alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_q)$ for q groups.

For each group g treated at time g and observed at times $g, g + 1, \dots, p$, we have at least $(p - g + 1)$ observations of θ_{gt} .

With ≥ 2 observations per group, the system is overdetermined and γ is uniquely identified via least squares, provided the design matrix has full rank.

Formally: the design matrix X with rows $(1, 0, \dots, 0, t - g, 0, \dots, 0)$ (indicator for group g in α positions, $t - g$ in β_g position) has rank $2q$ provided each group has ≥ 2 time periods and groups span different treatment times.

Therefore the mapping is injective. $\square$

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